CWR Lab Agrometeorology Documentation

GHG Flux Measurement Systems & Processing

CWR Lab Team

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1. Overview

Welcome to the CWR Lab micrometeorological measurement and analysis documentation. This resource is designed for new students, researchers, and staff working with CO2 and N2O flux measurement systems.

1.1 Measurement Sites

CWR Lab agrometeorology network include long-term benchmark sites and field towers across Canada. These sites span diverse climates and soil types from Ontario's humid croplands to the prairie provinces' dry continental systems, and support side-by-side comparisons of fertilizer practices, crop rotations, cover crops, and inhibitors. Each benchmark site combines continuous micrometeorological flux measurements with soil process studies, providing region-specific insights into nitrous oxide dynamics and the effectiveness of best management practices. Together, these sites form the foundation for scaling farm-level results into national strategies for emission reduction.

All network sites employ two complementary micrometeorological measurement techniques for quantifyin greenhouse gas and energy fluxes:

1. **Flux-Gradient (FG) Method** - Uses vertical concentration gradients
2. **Eddy Covariance (EC) Method** - Direct turbulent flux measurements

1.2 Quick Navigation

Introduction

About the ON1 site, instrumentation, and measurement principles

→ Get started

Eddy-Covariance

Direct turbulent flux measurements with high-frequency data

→ Learn EC

Flux-Gradient

Gradient-based flux estimation using MOST theory

→ Learn FG

Glossary

Complete variable definitions and diagnostic parameters

→ View glossary

1.3 Key Features

- **30-minute averaging intervals** for all diagnostic variables
- **High-frequency measurements** at 10 Hz (EC system)

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- **Dual-method comparison** for data validation
- **Comprehensive QA/QC guidelines**

1.4 Getting Started

New Users Start Here

1. Read the Site Overview to understand the measurement setup
2. Review the Instrumentation section
3. Choose measurement method: EC or FG
4. Familiarize with Quality Control procedures

1.5 Video Tutorial

Below is an introductory video explaining the site setup (example of video embedding):

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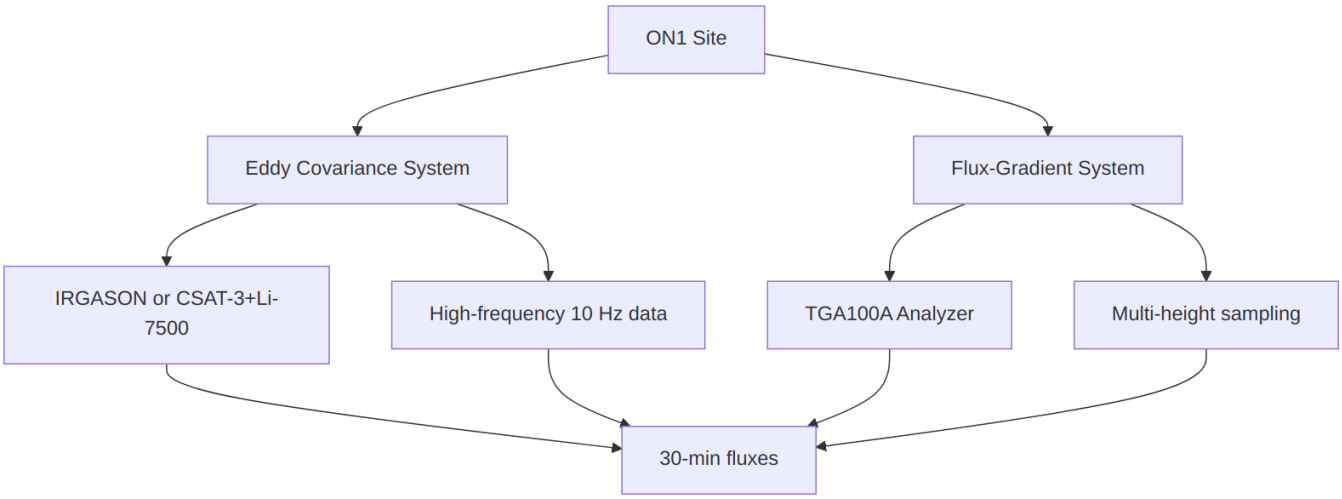
1.6 Site Diagram



1.7 Recent Updates

Date	Update
Dec 2025	Initial documentation compiled from instrument manuals
TBD	Site-specific calibration values
TBD	Measurement heights and footprint analysis

Important Note

The EC system at ON1 can be configured with either an **IRGASON** (integrated sensor) or separate **CSAT-3 + Li-7500** instruments. [PS: Verify the current configuration]

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2. Introduction

2.1 Overview

2.1.1 Site Overview

The main long-term site at Elora, Ontario (ON1) employs two complementary micrometeorological measurement techniques for quantifying greenhouse gas and energy fluxes. Understanding both methods allows for cross-validation and comprehensive flux characterization.

1. FLUX-GRADIENT (FG) METHOD

The flux-gradient method uses **vertical concentration gradients** to estimate fluxes. This semi-empirical approach requires parameterization of turbulent exchange through eddy diffusivity.

Key characteristics:

- Measures concentration differences at multiple heights
- Applies Monin-Obukhov Similarity Theory (MOST)
- More compact footprint than EC
- Better suited for heterogeneous surfaces
- Requires stable atmospheric conditions

2. EDDY COVARIANCE (EC) METHOD

The eddy covariance method provides **direct turbulent flux measurements** by correlating high-frequency fluctuations in wind and scalar concentrations.

Key characteristics:

- Direct flux measurement (no assumptions about turbulent transfer)
- High-frequency data collection (10 Hz)
- Widely accepted as standard method
- Larger spatial footprint
- Works in various stability conditions

2.1.2 Why Two Methods?

Complementary Approaches

Using both FG and EC methods provides:

- **Validation**: Agreement within 20-30% indicates well-functioning systems
- **Redundancy**: Backup measurements if one system fails
- **Different footprints**: Better spatial coverage
- **Method comparison**: Research and method development opportunities

2.1.3 Data Collection

All diagnostic variables are computed at **30-minute intervals** from high-frequency raw measurements.

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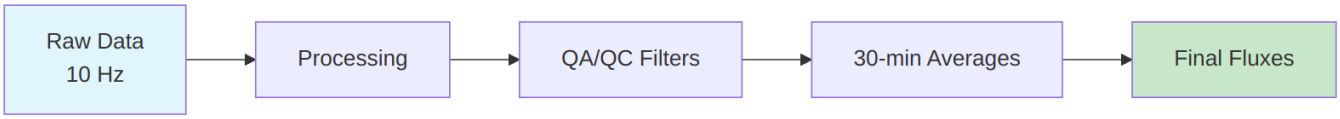

2.1.4 Measurement Frequency

System	Raw Frequency	Output Interval	Variables
EC System	10 Hz	30 minutes	CO_2, H_2O, momentum, heat
FG System	10 Hz	30 minutes	N_2O, CO_2, gradients

2.1.5 Site Photographs

Tower Setup *Figure 1: ON1 tower showing instrument placement*

EC Sensor *EC sensor configuration*

TGA Analyzer *TGA100A analyzer system*

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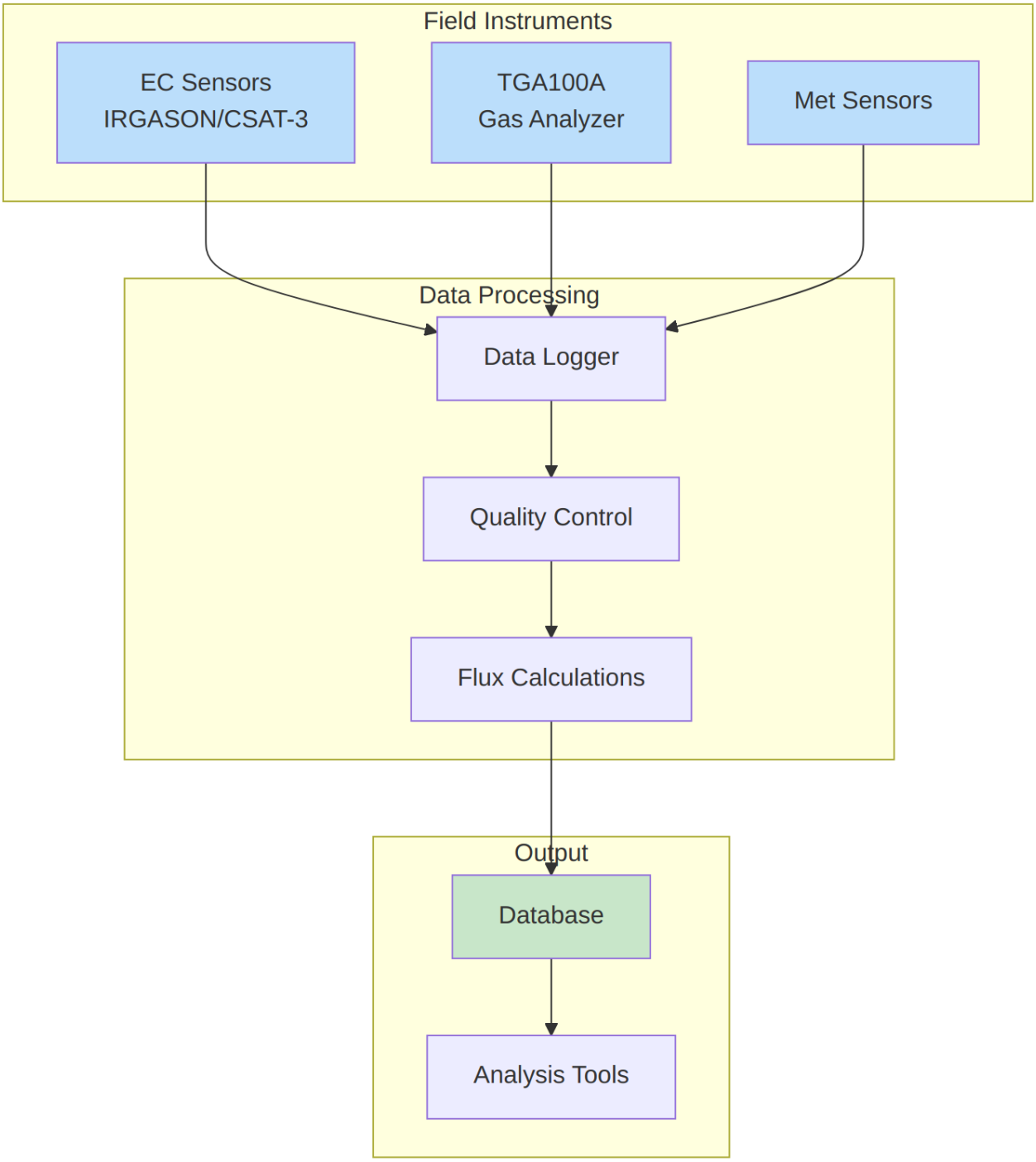
2.1.6 Data Flow Diagram



2.1.7 Timeline and Workflow

For new users, we recommend the following workflow:

1. **Week 1**: Read all documentation and instrument manuals
2. **Week 2**: Shadow experienced user during site visit
3. **Week 3**: Perform supervised data collection
4. **Week 4+**: Independent operation with periodic check-ins

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2.1.8 Key Contacts

Getting Help

- **Site Manager**: [Contact details]
- **Lab Supervisor**: [Contact details]
- **Technical Support**: [Contact details]
- **Emergency Contact**: [Contact details]

2.1.9 Next Steps

Continue to the detailed Site Description to learn about the physical setup and environmental context.

2.1.10 Video: Introduction to Flux Measurement

> 📹 *Video: Basic principles of flux measurements (15 min) — available in the online documentation at the project site.*

Video 1: Basic principles of flux measurements (15 minutes)

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2.2 Site Description

2.2.1 Location and Setting

Site Coordinates

- **Site Code**: ON1
- **Location**: [To be added]
- **Coordinates**: [Latitude, Longitude]
- **Elevation**: [meters above sea level]
- **Time Zone**: [Local time zone]

2.2.2 Land Use and Vegetation

Current Land Use

[Description of current agricultural or natural land use at the site]

Vegetation Characteristics

Parameter	Value/Description
Dominant species	[List main plant species]
Canopy height (hc)	[meters]
Leaf Area Index (LAI)	[m^2/m^2]
Growing season	[Months/dates]
Management	[Tillage, fertilization, irrigation]

Seasonal Variation

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Spring **Summer** **Fall** **Winter**

- Crop emergence
- Typical canopy height: [height]
- LAI development stage

- Peak growth period
- Maximum canopy height: [height]
- Maximum LAI: [value]

- Harvest period
- Senescence stage
- Crop residue management

- Bare soil or cover crop
- Minimal vegetation
- Potential snow cover

2.2.3 Climate

Annual Climate Summary

Parameter	Value	Notes
Mean annual temperature	[°C]	
Mean annual precipitation	[mm]	
Growing degree days	[GDD]	Base 5°C
Frost-free period	[days]	
Prevailing wind direction	[direction]	

Climate Normals (1991-2020)

Climate Graph *Figure: Monthly temperature and precipitation normals*

2.2.4 Soil Characteristics

Soil Type and Classification

- **Soil Order**: [Classification]
- **Texture**: [Clay/Loam/Sand percentages]
- **Drainage**: [Well/Moderately/Poorly drained]
- **pH**: [value]
- **Organic matter content**: [%]

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Soil Profile

Horizon	Depth (cm)	Description
A	0-20	[Description]
B	20-60	[Description]
C	60+	[Description]

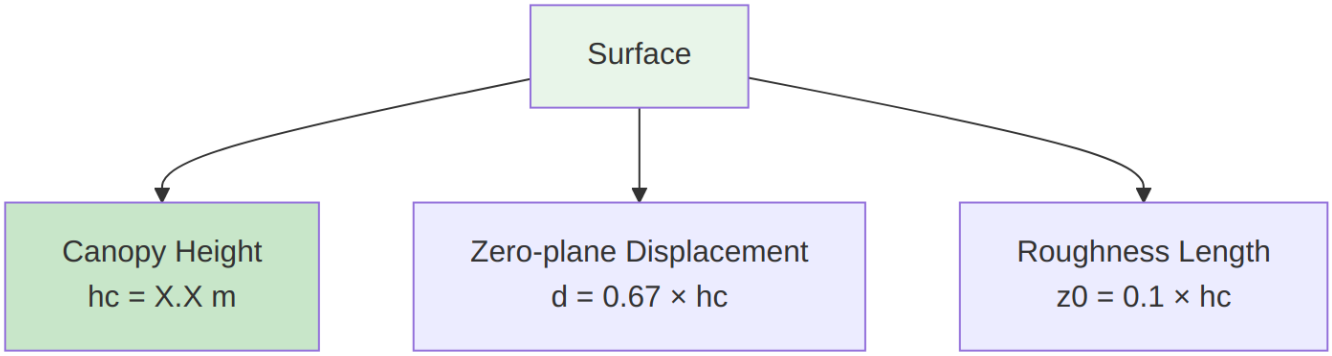
2.2.5 Tower Configuration

Measurement Heights

Tower Diagram *Figure: Vertical profile of instrumentation*

Instrument/Sensor	Height (m)	Purpose
EC sensors (IRGASON/CSAT-3)	[height]	Turbulent fluxes
FG upper intake	[height]	Concentration C_2
FG lower intake	[height]	Concentration C_1
Air temperature	[heights]	Profile measurements
Wind speed	[heights]	Profile measurements

Surface Roughness Parameters



Parameter	Symbol	Value	Notes
Canopy height	hc	[m]	Seasonal variation
Displacement height	d	[m]	~0.67 × hc
Roughness length	z_0	[m]	~0.1 × hc
Measurement height	zm	[m]	EC sensors

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2.2.6 Flux Footprint

Typical Footprint Characteristics

The spatial area contributing to flux measurements varies with:

- **Atmospheric stability**: Larger under stable, smaller under unstable
- **Wind speed**: Scales with mean wind
- **Measurement height**: Linear scaling
- **Surface roughness**: Affects near-field contribution

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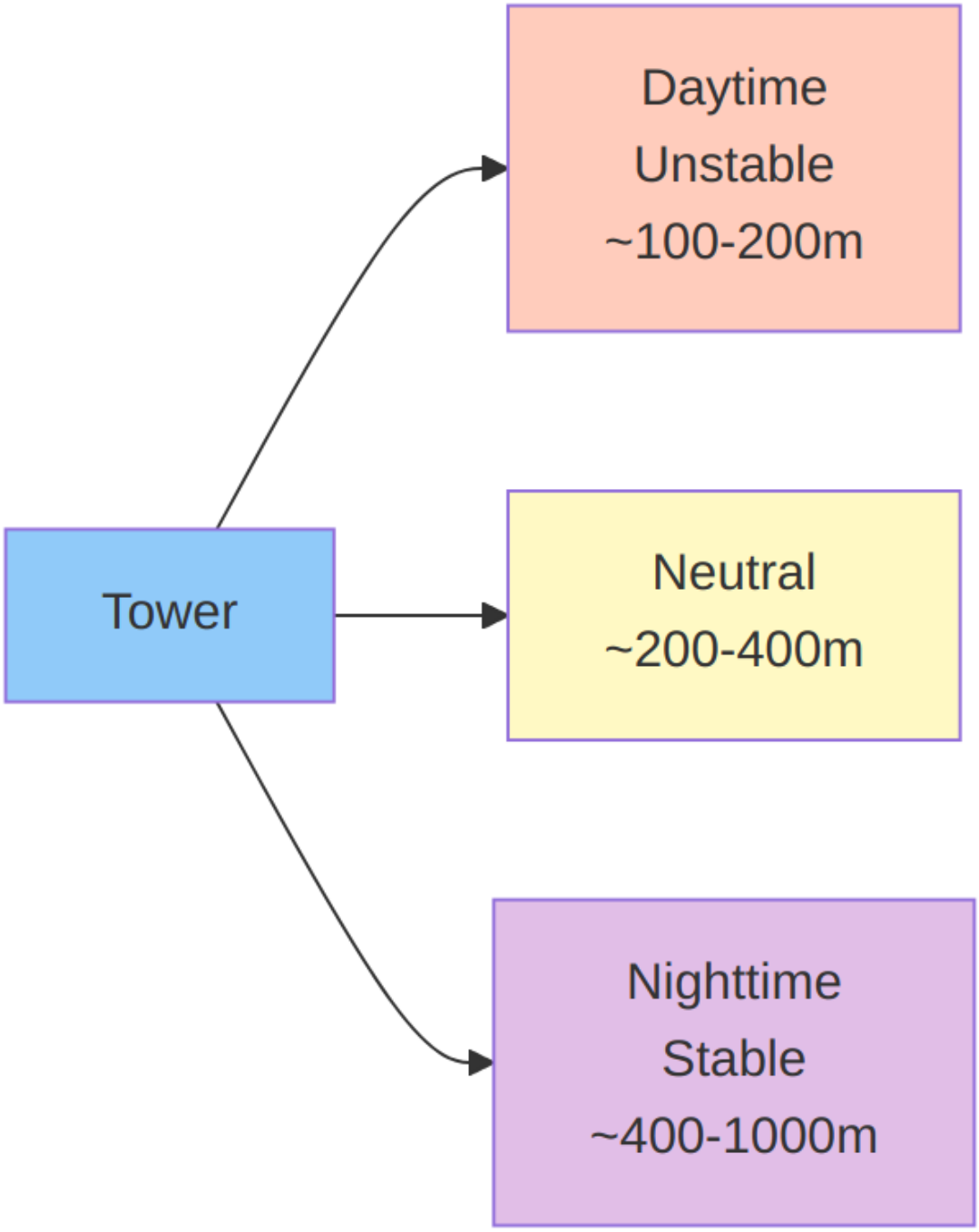
Footprint Analysis



Typical footprint distances:

Condition	Peak contribution	80% cumulative	90% cumulative
Unstable	~50m	~150m	~300m
Neutral	~100m	~300m	~600m
Stable	~200m	~600m	~1200m

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Site Homogeneity

Fetch Requirements

For representative measurements, the footprint area should be:

- ✓ Homogeneous in land cover
- ✓ Uniform in management
- ✓ Representative of target ecosystem
- ✗ Avoid roads, buildings, different crops

2.2.7 Surrounding Environment

Land Use Map

Site Map *Figure: Aerial view showing tower location and surrounding land use*

Nearby Features

Direction	Distance	Feature	Impact on measurements
North	[m]	[Feature]	[Potential influence]
East	[m]	[Feature]	[Potential influence]
South	[m]	[Feature]	[Potential influence]
West	[m]	[Feature]	[Potential influence]

2.2.8 Typical Flux Magnitudes

Expected Flux Ranges

Based on similar sites and preliminary data:

Flux	Growing Season	Non-Growing Season	Units
CO_2 (daytime)	-5 to -20	-2 to -5	μmol/m²/s
CO_2 (nighttime)	+2 to +10	+1 to +3	μmol/m²/s
N_2O	0 to +50	0 to +20	ng/m²/s
H (sensible heat)	50 to 300	-50 to 100	W/m²
LE (latent heat)	100 to 500	0 to 100	W/m²

Site-Specific Values

These ranges will be refined as more data is collected. Actual values depend on:

- Crop type and growth stage
- Management practices
- Weather conditions
- Soil moisture

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2.2.9 Access and Logistics

Site Access

- **Access road**: [Type and condition]
- **Parking**: [Location and capacity]
- **Walking distance**: [meters from parking to tower]
- **Terrain**: [Flat/sloped/difficult]

Safety Considerations

Safety Requirements

- [ ] Tower climbing certification required
- [ ] Two-person minimum for field work
- [ ] Weather check before site visit
- [ ] Emergency communication device
- [ ] First aid kit available
- [ ] Site emergency plan reviewed

Equipment Storage

- **Instrument shelter**: [Description]
- **Data logger**: [Location]
- **Calibration gases**: [Storage location]
- **Maintenance tools**: [Storage location]

2.2.10 Data Management

Data Logger Configuration

Parameter	Value
Logger model	Campbell Scientific [model]
Sampling frequency	10 Hz (EC), 30 min (output)
Storage capacity	[GB]
Communication	[Cellular/satellite/manual]
Data backup	[Frequency and method]

Data Transfer

- **Real-time**: [If applicable]
- **Manual download**: [Frequency]
- **Backup schedule**: [Weekly/monthly]
- **Quality control**: [Who and when]

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2.2.11 Historical Context

Site History

Period	Land Use	Notes
Pre-2010	[Previous use]	
2010-2020	[Land use]	
2020-Present	[Current use]	

Measurement History

Year	Activity	Notes
[Year]	Site established	Initial setup
[Year]	EC system installed	[Configuration]
[Year]	FG system added	[Configuration]

2.2.12 Contact and Responsibilities

Site Management

Role	Name	Contact	Responsibilities
Site Manager	[Name]	[Email/Phone]	Overall operations
Data Manager	[Name]	[Email/Phone]	QA/QC, database
Field Tech	[Name]	[Email/Phone]	Maintenance
PI	[Name]	[Email/Phone]	Scientific oversight

Emergency Contacts

Type	Contact	Phone
Site Manager	[Name]	[Number]
Lab Supervisor	[Name]	[Number]
Emergency Services	911	
Landowner	[Name]	[Number]

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2.2.13 Photo Gallery

Tower Overview

Tower Overview

View from south showing instrument placement

EC Sensors

EC Sensors

IRGASON mounted at measurement height

TGA System

TGA System

Interior showing TGA100A installation

Site Context

Aerial View

Tower location within field

2.2.14 Next Steps

Now that you understand the site context:

- Learn about Instrumentation
- Study EC Method
- Study FG Method

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2.3 Instrumentation

2.3.1 System Overview

The ON1 site utilizes state-of-the-art micrometeorological instruments from Campbell Scientific and LI-COR Biosciences to measure greenhouse gas and energy fluxes.

2.3.2 Flux-Gradient System

Primary Instrument: TGA100A Trace Gas Analyzer

The Campbell Scientific TGA100A is the heart of the flux-gradient measurement system.

TGA100A System

TECHNOLOGY

- **Measurement principle**: Tunable diode laser absorption spectroscopy
- **Measurement frequency**: 10 Hz
- **Target gases**: N_2O and CO_2
- **Path length**: 153.08 cm (sample cell)
- **Cooling system**: Liquid nitrogen (LN_2) for laser stabilization

MEASUREMENT APPROACH

Multi-plot flux-gradient using dual-ramp absorption spectroscopy enables precise concentration measurements at multiple sampling heights.

TGA100A Specifications

Parameter	Specification
Sample cell path	153.08 cm
Operating pressure	50-80 mb
Sample flow	500-2000 ml/min
Measurement frequency	10 Hz
Precision (N_2O)	< 0.3 ppb
Precision (CO_2)	< 0.1 ppm

Supporting Components

The TGA system requires several supporting components for proper operation:

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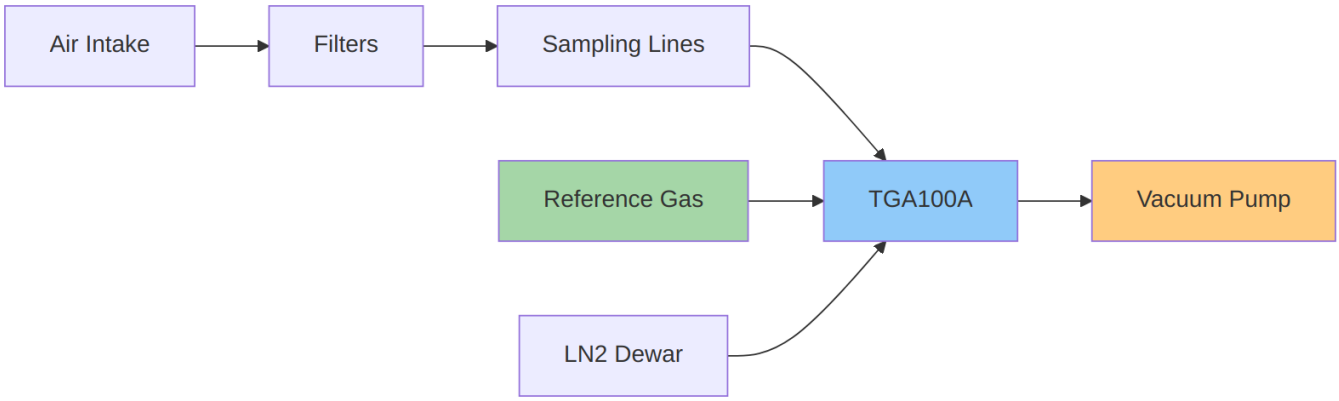

- **Sampling system**: Multi-height intake with automated switching
- **Vacuum pump**: Maintains controlled sample pressure
- **Reference gas supply**: For calibration and drift correction
- **Temperature and pressure sensors**: For gas density corrections
- **Automated calibration system**: Periodic zero and span checks

2.3.3 Eddy Covariance System

The ON1 site can be configured with one of two EC system setups. Verify the current configuration.

Configuration 1: IRGASON

IRGASON Sensor

Integrated CO_2/H_2O Open-Path Gas Analyzer and 3-D Sonic Anemometer

The IRGASON combines gas analysis and wind measurement in a single, co-located instrument, minimizing flux loss from sensor separation.

SPECIFICATIONS

Parameter	Value
Manufacturer	Campbell Scientific
Maximum measurement rate	60 Hz
User-programmable output	5, 10, 12.5, or 20 Hz
Gas analyzer path length	15.37 cm
CO_2 calibrated range	0-1000 µmol/mol
H_2O calibrated range	0-72 mmol/mol

ADVANTAGES

- Co-located measurements reduce flux loss
- Simplified installation
- Single power supply and data connection
- Integrated heating for cold weather operation

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Configuration 2: CSAT-3 + Li-7500

COMPONENT A: CSAT-3 SONIC ANEMOMETER

CSAT-3

Three-Dimensional Sonic Anemometer

Measures wind velocity components and sonic temperature at high frequency.

Key Specifications:

- **Manufacturer**: Campbell Scientific
- **Measurement frequency**: 60 Hz (typical operation at 10 or 20 Hz)
- **Measured variables**:
  - 3-D wind components (Ux, Uy, Uz)
  - Sonic temperature (Ts)
- **Wind offset specification**:
  - Horizontal: < ±8.0 cm/s
  - Vertical: < ±4.0 cm/s

Coordinate System

The CSAT-3 measures wind in its own coordinate system, which requires rotation to align with mean wind direction (coordinate rotation is part of standard EC processing).

COMPONENT B: LI-7500 OPEN-PATH ANALYZER

Li-7500

Open-Path CO_2/H_2O Gas Analyzer

Measures CO_2 and H_2O concentrations using infrared absorption.

Key Specifications:

- **Manufacturer**: LI-COR Biosciences
- **Measurement bandwidth**: 5, 10, or 20 Hz (software selectable)
- **Path length**: 12.5 cm
- **Precision (RMS)**:
  - CO_2: 0.2 mg/m³
  - H_2O: 0.004 g/m³

Open-Path Considerations

Open-path analyzers are affected by:

- Rain and fog (contamination)
- Temperature fluctuations (WPL corrections required)
- Dust and debris accumulation

Regular cleaning and maintenance are essential.

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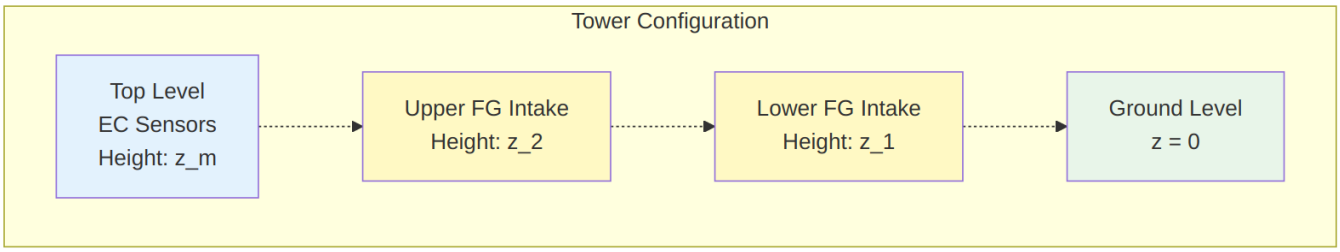
2.3.4 Sensor Placement and Geometry



Typical Heights (Site-Specific)

Example Configuration

Level	Height (m)	Measurement
EC sensors	3.0	Direct fluxes
FG upper intake	2.5	Concentration C_2
FG lower intake	0.5	Concentration C_1
Canopy height	0.3	Reference level

2.3.5 Maintenance Schedule

Daily Checks

- [ ] TGA sample pressure and flow
- [ ] EC data quality flags
- [ ] Visual inspection of sensors
- [ ] Check data logger connection

Weekly Checks

- [ ] Clean EC sensor windows
- [ ] Check TGA LN_2 level
- [ ] Inspect sampling lines for leaks
- [ ] Review data quality metrics

Monthly Checks

- [ ] Full system calibration
- [ ] Replace inlet filters
- [ ] Check all electrical connections
- [ ] Backup data and logs

Quarterly Maintenance

- [ ] Professional calibration service
- [ ] Replace consumables
- [ ] Detailed performance review

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- [ ] Update site documentation

2.3.6 Instrument Manuals

All detailed instrument manuals are available in the project files:

- **Campbell TGA100A Manual**
- **Campbell CSAT-3 Manual**
- **LI-COR Li-7500 Manual**

> **Quick Reference**
>
> Bookmark the following manual sections for quick access:
>
> - **TGA100A**: Section 7.1.2 (Routine System Checks)
> - **CSAT-3**: Section 11.2.2 (Wind Offset Testing)
> - **Li-7500**: Section 5 (Maintenance)

2.3.7 Video Demonstrations

TGA System Setup

> 📹 *Video demonstration — see online documentation — available in the online documentation at the project site.*

Video: TGA100A setup and calibration procedure (20 minutes)

EC Sensor Installation

> 📹 *Video demonstration — see online documentation — available in the online documentation at the project site.*

Video: EC sensor mounting and alignment (15 minutes)

2.3.8 Next Steps

Now that you understand the instrumentation, proceed to learn about the specific measurement methods:

- Eddy-Covariance Fundamentals
- Flux-Gradient Fundamentals

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3. Processing

3.1 Overview of the Processing Workflow

This section provides a high-level overview of the end-to-end data-processing workflow used in the CanN2Onet project, from field sensors to archived flux products.

The goal is to document **how data flow through the system**, not to re-derive eddy-covariance or flux-gradient theory (which is available in standard references).

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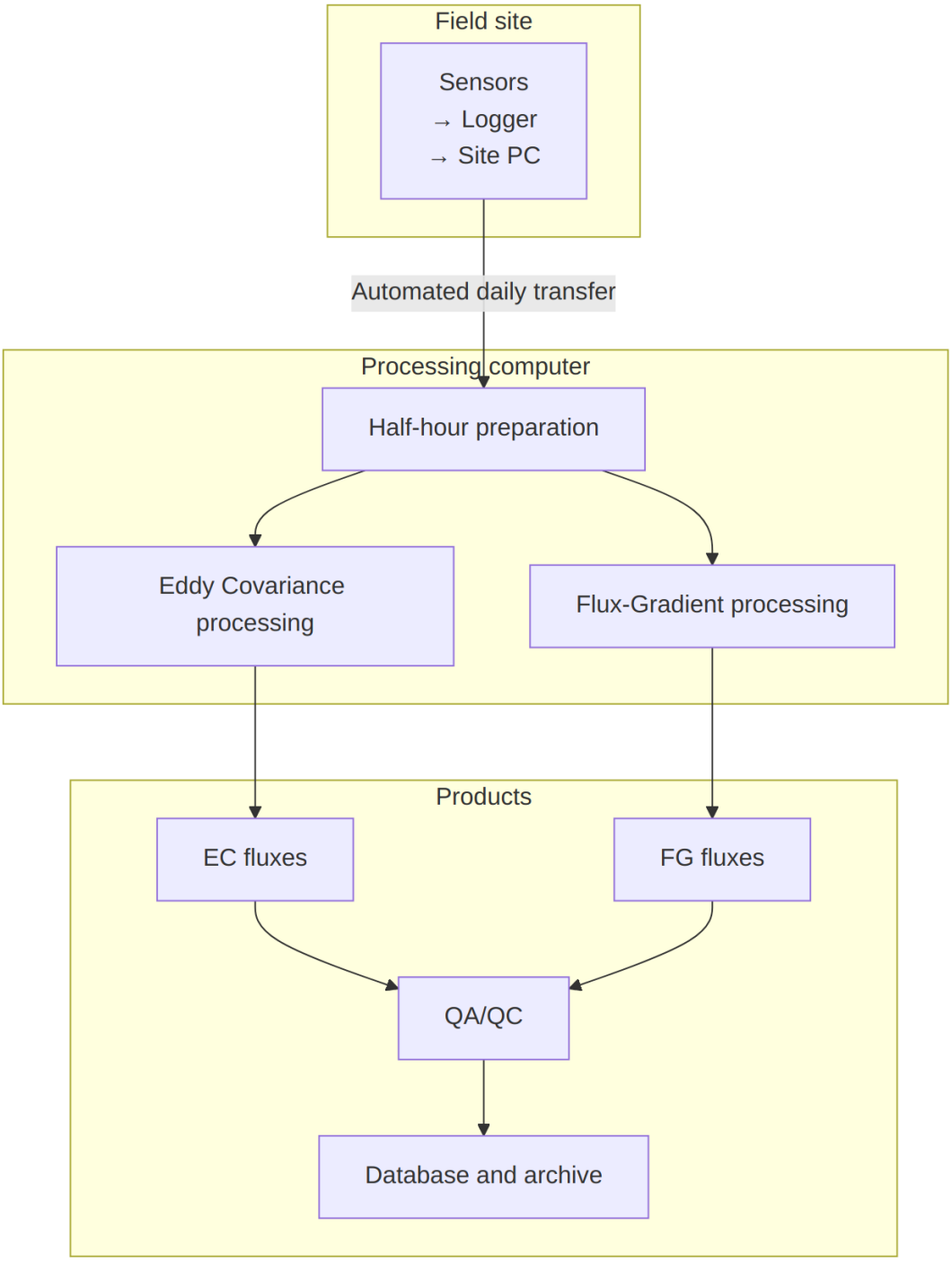
3.1.1 Conceptual data flow



3.1.2 Key characteristics

- Fully automated daily processing on a Windows workstation
- Combination of Python, MATLAB, and EddyPro workflows
- Site-specific configuration via initialization scripts
- ASCII vector outputs used as an intermediate archival format

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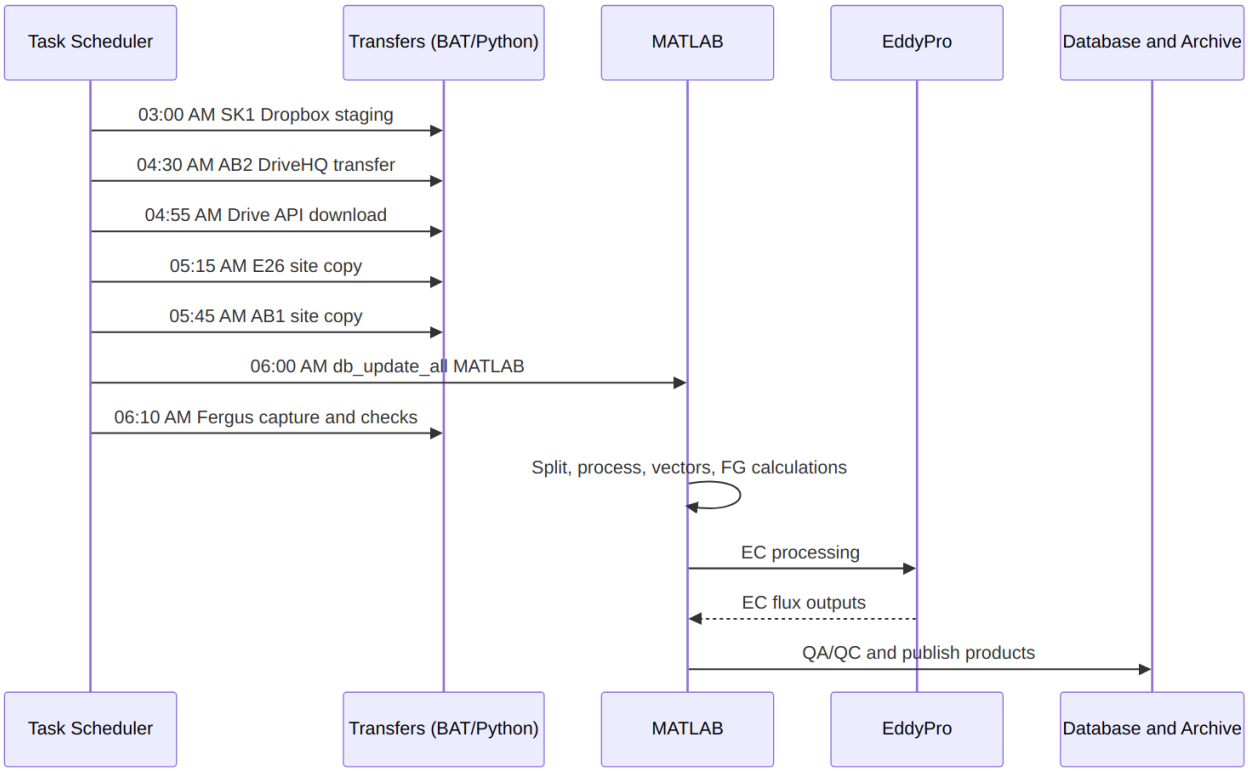
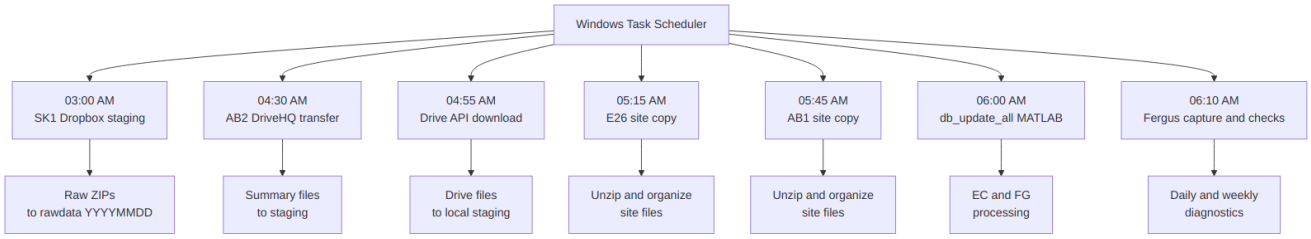
3.2 Automation and Task Scheduler

Daily processing is orchestrated using **Windows Task Scheduler**. Each scheduled task launches a batch (`.bat`) file that in turn calls Python or MATLAB scripts.

3.2.1 Scheduled Daily Tasks Overview



Tasks At-a-Glance



3.2.2 Design rationale

- Data ingestion is separated from scientific processing
- Failures in one site do not block others
- Logs are written for traceability and recovery

Data Transfer Details

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For detailed information on how data are transferred from field sites (Google Drive API, Dropbox, DriveHQ, and hybrid systems), including workflows and pseudocode, see:

➤ **Data Transfer Workflows**

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3.3 Data Transfer Workflows

3.3.1 Overview

This document describes how field data are transferred from site computers to the lab processing system across all CanN2Onet sites.

While transfer mechanisms vary (Google Drive API, Dropbox, DriveHQ), all sites follow a common logical workflow:

```
Site Computer → Transfer Mechanism → Lab Staging → Unpack/Organize → Archive → Processing
```

This document provides:

- a high-level architecture
- a common workflow (pseudocode)
- transfer mechanism patterns
- a comparison table across sites
- site-specific implementations

3.3.2 1. Data Transfer Architecture

Different sites use different transfer mechanisms depending on:

- field connectivity
- infrastructure constraints
- historical implementation

Supported Transfer Mechanisms

- Google Drive API (scripted upload/download)
- Dropbox synchronization (passive sync)
- DriveHQ / FTP-style cloud services
- Hybrid approaches

Key Design Principles

- Data ingestion is separated from processing
- Transfer is fault-tolerant and restartable
- Files are archived after transfer
- Temporary staging is used before processing

3.3.3 2. Common Workflow (Pseudocode)

```
ALGORITHM DATA_TRANSFER_WORKFLOW

INPUT:
    site-generated data files (tower, TGA, manual)
    transfer mechanism (API / Dropbox / DriveHQ)

OUTPUT:
    organized data in lab repository
    archived transfer files

BEGIN
```


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```
1. DETECT new data files
    - via API listing, folder scan, or sync client

2. TRANSFER to lab staging
    - download or sync into local staging directory

3. FOR each transferred file:

    IF file is compressed (ZIP):
        copy to temporary directory
        unzip into destination folder

    DETERMINE destination based on:
        - data type (TGA, EC, manual, etc.)
        - site configuration
        - date extracted from filename

    COPY data to:
        (a) main repository
        (b) site mirror directory (if applicable)

    IF file requires conversion (e.g., RawData):
        run conversion utility
        store converted output

    MOVE original transfer file to archive location

    VERIFY transfer success (e.g., file size match)

    IF verification successful:
        remove temporary/local copy
        remove remote copy (if API-based)

END FOR

END
```

3.3.4 3. Transfer Mechanism Patterns

3.1 Google Drive API

OVERVIEW

- Used by: ON1, ON3, AB1, AB2 (partial), SK1 (manual data)
- Fully scripted upload and download

WORKFLOW

```
Site → Local "files" folder → Google Drive (API upload)
     → Lab download script → Local staging → Processing
```

IMPLEMENTATION DETAILS

- Authentication via service account (`wagner-riddle-auth.json`)
- Upload:
- files copied into local `files/` directory
- uploaded using Drive API (`files().create`)
- Download:
- list files using `files().list()`
- download via `MediaIoBaseDownload`
- Cleanup:
- delete from Drive after successful processing
- Acts as **temporary staging**, not long-term storage

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3.2 Dropbox Sync

OVERVIEW

- Used by: SK1 (tower + TGA)
- Passive synchronization (no upload script)

WORKFLOW

```
Site Dropbox folder → Cloud sync → Lab Dropbox folder → Processing
```

IMPLEMENTATION DETAILS

- Site computer:
- Dropbox client monitors directories
- automatic upload (no scripting)
- Lab computer:
- Dropbox syncs files locally
- Differences from API:
- no explicit upload/download control
- timing depends on sync completion
- no automatic deletion after processing

3.3 DriveHQ / FTP-style Transfer

OVERVIEW

- Used by: AB2 (tower data)
- External cloud/FTP-style system

WORKFLOW

```
Site → DriveHQ upload → Lab retrieval script → Processing
```

IMPLEMENTATION DETAILS

- Upload handled externally (not in current scripts)
- Lab side:
- scheduled retrieval/download
- Differences:
- separate authentication system
- less integrated with pipeline
- typically used for tower datasets

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3.3.5 4. Site Comparison Table

Site	Tower Data	TGA Data	Manual / Heights	Method Type
ON1	Google API	Google API	Google API	Fully API
ON3	Google API	Google API	Google API	Fully API
SK1	Dropbox	Dropbox	Google API	Hybrid
AB1	External / local + scripts	Google API	Google API	Hybrid
AB2	DriveHQ	Google API	Google API	Hybrid

3.3.6 5. Site-Specific Implementations

5.1 ON1 – Google Drive API Transfer

OVERVIEW

- Transfer method: Google Drive API
- Data: tower, TGA, manual, logbook
- Automation: Python + Task Scheduler

WORKFLOW SUMMARY

- Site: collect → rename → zip → upload
- Lab: download → unzip → organize → archive → cleanup

DESTINATIONS

Main repository

```
ses-lab-data\E26\hfdata\...
ses-lab-data\E26\database\...
```

Mirror copy

```
ses-lab-data\Guelph_agromet\sites\ON1\...
```

Transfer archive

```
ses-lab-data\Transfers\E26\...
```

NOTES

- Files copied to both main repository and mirror
- ZIP files archived after processing
- Files removed from Drive only after verification

5.2 ON3 – Google Drive API Transfer

OVERVIEW

- Same architecture as ON1
- Different site configuration and paths

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DIFFERENCES FROM ON1

- station naming
- directory structure
- fewer/more instruments

5.3 SK1 – Dropbox + Google API (Hybrid)

OVERVIEW

- Tower + TGA: Dropbox
- Manual data: Google API

WORKFLOW SUMMARY

- Dropbox handles high-frequency data
- API handles manual/logbook files

KEY DIFFERENCES

- No upload script for tower data
- Sync timing depends on Dropbox
- No automatic deletion of remote files

5.4 AB1 – Hybrid Transfer

OVERVIEW

- Uses Google API for TGA + manual data
- Additional scripts for SmartFlux / summary data

NOTES

- Multiple independent ingestion paths
- Requires coordination at processing stage

5.5 AB2 – DriveHQ + Google API (Hybrid)

OVERVIEW

- Tower data: DriveHQ
- TGA + manual: Google API

WORKFLOW SUMMARY

- DriveHQ used for tower (SmartFlux) datasets
- API used for TGA and other structured datasets

KEY DIFFERENCES

- External cloud replaces site upload script
- Lab retrieval handled separately
- Integrated downstream with same processing pipeline

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3.3.7 Summary

Despite different transfer mechanisms, all sites converge into a unified workflow:

```
Transfer → Staging → Organization → Archive → Processing
```

This design allows flexibility at the site level while maintaining consistency in downstream processing.

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3.4 Eddy Covariance Processing Pipeline

The **Eddy Covariance (EC)** branch of the processing system.

3.4.1 Scope & overview

- High-frequency (typically 10 Hz) EC data
- Half-hour segmentation
- Flux calculation using EddyPro
- Post-processing and filtering prior to archiving

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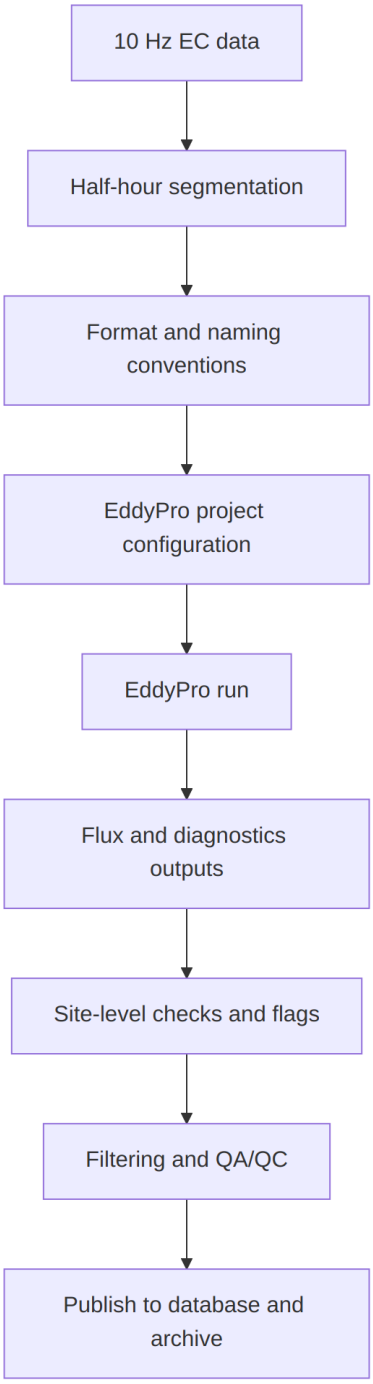

3.4.2 Notes

- Instrument-specific details (e.g., IRGASON, CSAT3, LI-7500) are handled upstream
- The EC workflow is largely external to the MATLAB db_* system
- Outputs are synchronized with FG products for unified archiving

Continue to the detailed Eddy-Covariance Analysis to learn about the setup and context.

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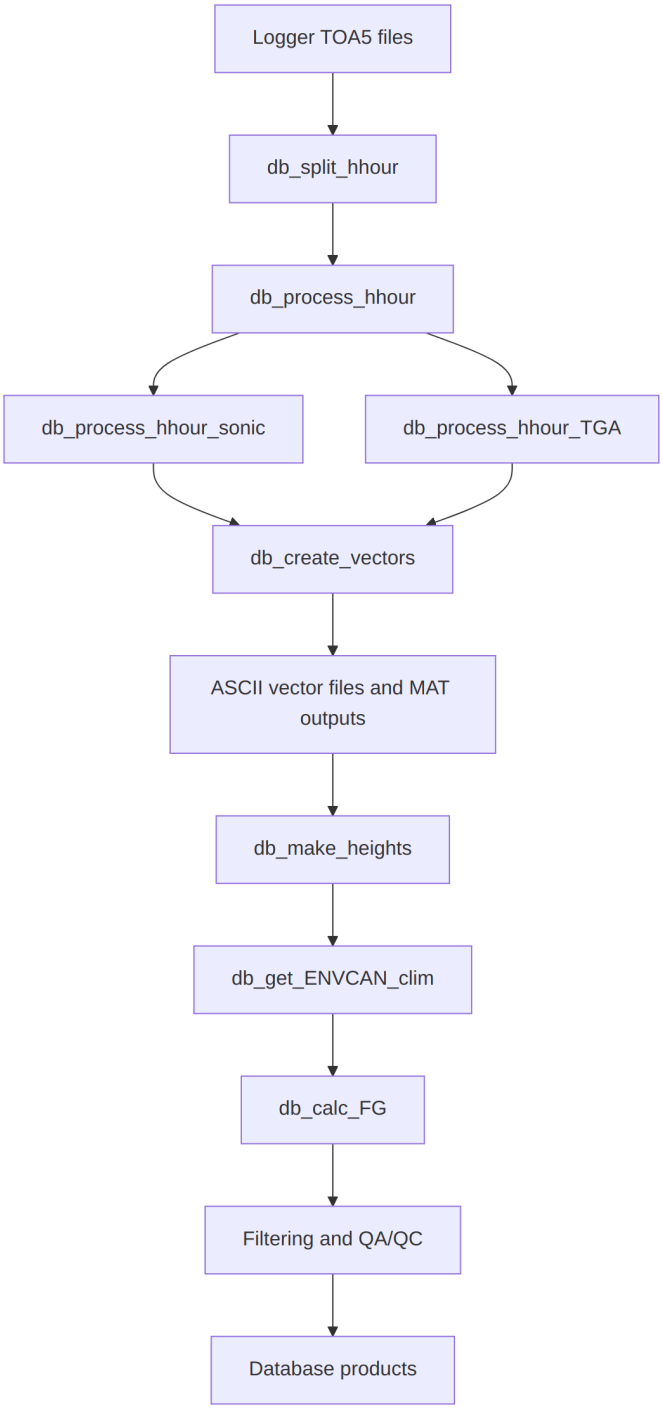
3.5 Flux-Gradient Processing Pipeline

The **Flux-Gradient (FG)** processing chain implemented primarily in MATLAB:



3.5.1 Notes

- The main entry point is `db_update_site.m`
- Site behavior is controlled by the site specific initialization script `<SITE>_init_all.m`
- MATLAB (`.mat`) structure files and ASCII output vector files are generated for diagnostics, sharing and subsequent archiving

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Continue to the detailed Flux-Gradient Analysis to learn about the setup and context.

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4. Eddy-Covariance Analysis

4.1 Eddy Covariance Fundamentals

4.1.1 What is Eddy Covariance?

The **Eddy Covariance (EC) method** directly measures turbulent fluxes by computing the covariance between high-frequency fluctuations in vertical wind velocity and the scalar of interest (e.g., CO_2, H_2O, temperature).

EC Concept *Figure: Eddy covariance measures turbulent transport by correlating vertical wind with scalar concentrations*

4.1.2 Theoretical Basis

Reynolds Decomposition

Any atmospheric variable can be decomposed into mean and fluctuating components:

$$c = \overline{c} + c'$$

where:

- c = instantaneous concentration
- $\overline{c}$ = time-averaged mean concentration
- c' = turbulent fluctuation from the mean

The Flux Equation

The vertical flux of a scalar is calculated as:

$$F_c = \overline{w'c'}$$

where:

- F_c = vertical flux of scalar c
- w' = vertical wind velocity fluctuation
- c' = scalar concentration fluctuation
- Overbar denotes time averaging

Why This Works

When upward air motion ($w'>0$) consistently carries higher concentrations ($c'>0$), the covariance $\overline{w'c'}$ is positive, indicating upward flux. The method captures the actual turbulent transport without requiring assumptions about the transfer mechanism.

4.1.3 Key Assumptions

The EC method relies on several assumptions:

1. Stationarity

Turbulent statistics should not change over the averaging period (typically 30 minutes).

Check: Divide 30-min period into sub-periods and verify similar statistics.

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2. Horizontal Homogeneity

Surface properties should be uniform in the flux footprint.

Implication: Site selection is critical for representative measurements.

3. Negligible Advection

Horizontal and vertical advection should be small compared to turbulent flux.

When violated: Over complex terrain or with strong mesoscale flows.

4. Complete Turbulent Sampling

The averaging period must capture all relevant turbulent eddies.

Typical requirement: 30-60 minute averaging periods.

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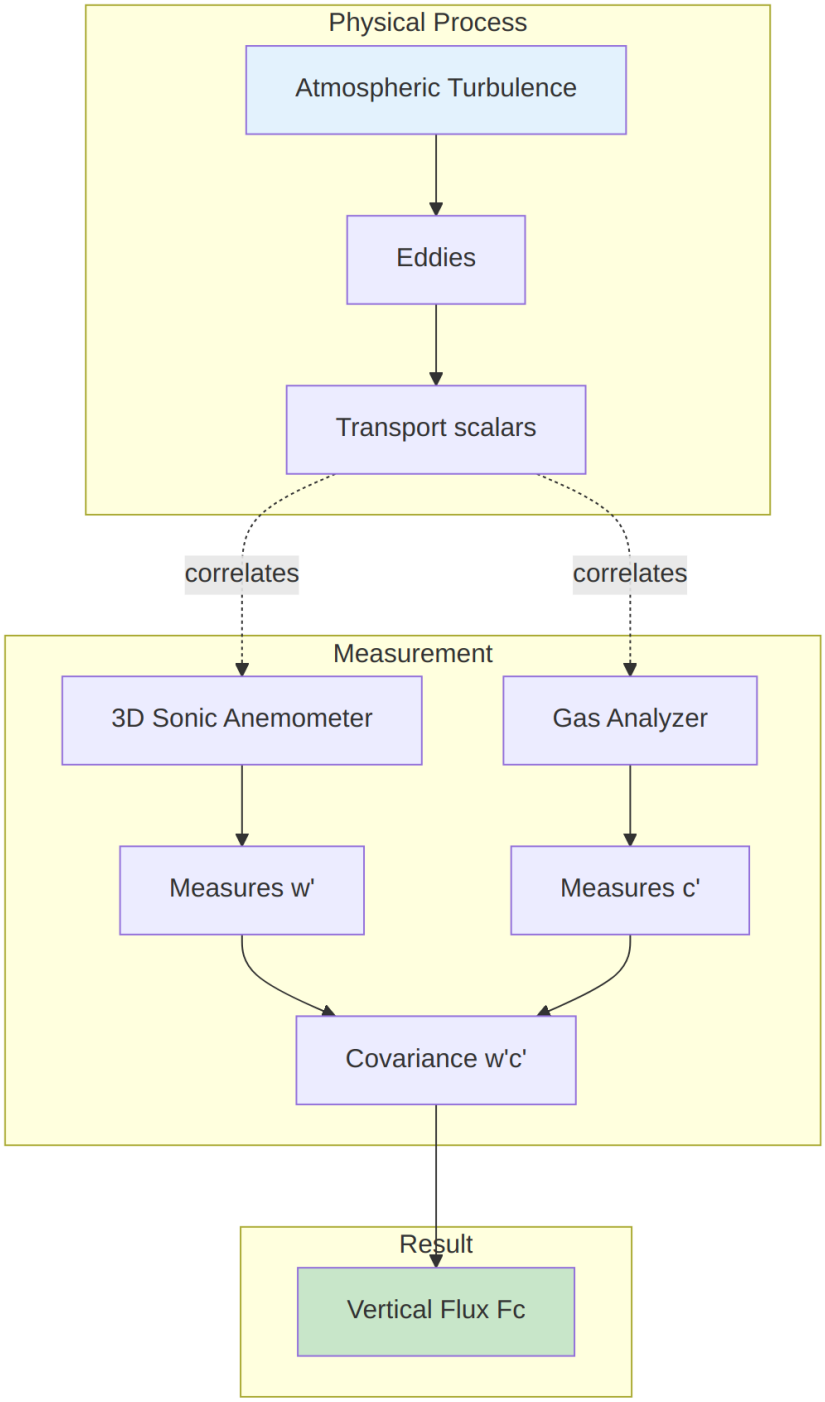
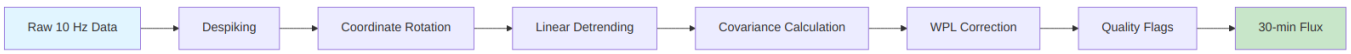
4.1.4 Measurement Principle Diagram



4.1.5 Flux Calculations

Basic Steps



1. COORDINATE ROTATION

Rotate coordinate system so mean vertical velocity is zero:

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- **Double rotation**: Most common method
- **Planar fit**: For complex terrain (requires long-term data)

2. LINEAR DETRENDING

Remove low-frequency trends that aren't turbulent:

$$x'(t)=x(t)-[a+bt]$$

3. COVARIANCE CALCULATION

For CO_2 flux:

$$F_{CO_2}=\overline{w'c'}=\frac{1}{N}\sum i=1Nw_i'\cdot c_i'$$

4. WPL CORRECTION

Webb-Pearman-Leuning (WPL) correction accounts for density effects:

$$F_{CO_2,corrected}=F_{CO_2,measured}+\mu\cdot\overline{c}\cdot E+(1+\mu\cdot\sigma)\cdot\overline{c}\cdot\frac{H}{\overline{T}}$$

where:

- E = water vapor flux
- H = sensible heat flux
- μ = ratio of molecular masses
- σ = ratio of molar densities

> ⚠ **Critical for Open-Path Sensors**
>
> WPL corrections are essential for open-path analyzers and can be 10-20% of measured flux.

4.1.6 Example Fluxes

CO_2 Flux

Daytime (Photosynthesis):

$$F_{CO_2}<0(\text{uptake})$$

Nighttime (Respiration):

$$F_{CO_2}>0(\text{emission})$$

Latent Heat Flux

Evapotranspiration from surface:

$$LE=\lambda\cdot E=\lambda\cdot\overline{w'\rho_v'}$$

where:

- λ = latent heat of vaporization (2.45 MJ/kg)
- E = water vapor flux
- ρ_v = water vapor density

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Sensible Heat Flux

$$H = \rho \cdot c_p \cdot \overline{w'T'}$$

where:

- ρ = air density
- c_p = specific heat of air (1005 J/kg/K)
- T = temperature

Momentum Flux (Friction Velocity)

$$u_* = [(\overline{u'w'})^2 + (\overline{v'w'})^2]^{1/4}$$

Friction velocity (u_*) characterizes turbulent intensity and is crucial for:

- Flux-gradient calculations
- Footprint modeling
- Quality control

4.1.7 Typical Values

Daily Cycle Example

Time	CO_2 Flux (μmol/m²/s)	H (W/m²)	LE (W/m²)	u* (m/s)
06:00	+2	-20	50	0.15
12:00	-15	250	300	0.45
18:00	+5	80	150	0.30
24:00	+8	-30	20	0.12

Sign Conventions

- **Negative flux**: Downward (into surface)
- **Positive flux**: Upward (from surface)

For CO_2: Negative = ecosystem uptake (photosynthesis)

4.1.8 Data Quality Considerations

High-Quality Data Requires:

✓ **Sufficient turbulence**: $u_* > 0.1$ m/s (threshold varies by site)
✓ **Stationarity**: Flux covariance quality test
✓ **Horizontal homogeneity**: Footprint within target area
✓ **Proper sensor function**: No spikes, offsets, or drift
✓ **Complete data**: < 10% missing high-frequency data

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Common Issues

Insufficient Turbulence **Sensor Malfunction** **Rain Events**

Problem: Stable atmospheric conditions (nighttime)
Solution: Use u_* threshold filtering

Problem: Dirty optics, calibration drift
Solution: Regular maintenance and calibration

Problem: Water on sensor windows
Solution: Automated detection and flagging

4.1.9 Interactive Visualization

> 📹 *Interactive figure: Typical daily flux patterns — see online documentation — available in the online documentation at the project site.*

Interactive Figure: Typical daily flux patterns (hover for details)

4.1.10 Video Lecture: EC Theory

> 📹 *Video: Eddy covariance theory and practice (30 min) — see online documentation — available in the online documentation at the project site.*

> 📹 *Interactive figure: Typical daily flux patterns — see online documentation — available in the online documentation at the project site.*

Video: Eddy covariance theory and practice (30 minutes)

4.1.11 Further Reading

Recommended Papers

1. **Baldocchi, D.D. (2014)**. "Measuring fluxes of trace gases and energy between ecosystems and the atmosphere." *Global Change Biology*, 20, 3600-3609. [Foundation paper]
2. **Webb et al. (1980)**. "Correction of flux measurements for density effects." *QJRMS*, 106, 85-100. [WPL correction]
3. **Moncrieff et al. (1997)**. "A system to measure surface fluxes." *Journal of Hydrology*, 188-189, 589-611. [Practical implementation]

4.1.12 Next Steps

- EC Variables - Detailed variable definitions
- Data Processing - Step-by-step processing workflow
- Quality Control - QA/QC procedures

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5. Flux-Gradient Analysis

5.1 Flux-Gradient Fundamentals

5.1.1 What is the Flux-Gradient Method?

The **Flux-Gradient (FG) method** estimates vertical fluxes by measuring concentration or temperature gradients and relating them to turbulent exchange through **Monin-Obukhov Similarity Theory (MOST)**.

The CWR Lab FG system — developed by the Wagner-Riddle group at the University of Guelph — measures **N_2O and CO_2** fluxes using the Campbell Scientific TGA100A and a dual-height intake arrangement on each measurement plot.

FG Concept *Figure: Flux-gradient method uses vertical gradients and turbulent diffusivity*

5.1.2 Physical Constants

The MATLAB pipeline uses the following physical constants, appropriate for the University of Guelph site location (~43.5 °N):

Constant	Symbol	Value	Units
von Kármán constant	κ	0.40	—
Gravitational acceleration	g	9.81	m s^{-2}
Universal gas constant	R	8.31451	J mol^{-1} K^{-1}
Specific gas constant (dry air)	R_d	287.05	J kg^{-1} K^{-1}
Molar mass of dry air	M_air	28.96	g mol^{-1}
Molar mass of CO_2	M_CO_2	44.01	g mol^{-1}
Molar mass of N_2O	M_N_2O	44.01	g mol^{-1}

Gravity at Guelph

g = 9.81 m s^{-2} is the appropriate value for ~43.5 °N latitude. Gravitational acceleration increases toward the poles due to Earth's oblate shape and rotation; using a geographically accurate value reduces systematic error in the Obukhov length calculation.

5.1.3 Theoretical Foundation

Basic Principle

Flux is proportional to the concentration gradient:

$$F_c = -K_c \frac{\partial c}{\partial z}$$

where:

- F_c = vertical flux of scalar c
- K_c = eddy diffusivity for scalar c (m² s^{-1})
- $\frac{\partial c}{\partial z}$ = vertical concentration gradient

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The Challenge

Unlike molecular diffusion where diffusivity is constant, **eddy diffusivity** K_c varies with atmospheric stability, wind speed, and measurement height. MOST provides the theoretical framework to determine K_c.

5.1.4 Monin-Obukhov Similarity Theory (MOST)

Obukhov Length

Atmospheric stability is characterised by the **Obukhov length** L:

$$L = -\frac{u_*^3 \cdot T_v}{\kappa \cdot g \cdot \overline{w'T'}}$$

where:

- u_* = friction velocity (m s⁻¹)
- T_v = virtual temperature (K) — the MATLAB pipeline uses virtual temperature (rather than sonic temperature alone) to apply a proper humidity correction
- κ = 0.40 (von Kármán constant)
- g = 9.81 m s⁻²
- $\overline{w'T'}$ = kinematic sensible heat flux (K m s⁻¹)

Stability Parameter

The dimensionless stability parameter:

$$\zeta = \frac{z-d}{L}$$

where z is measurement height (m) and d is zero-plane displacement (m).

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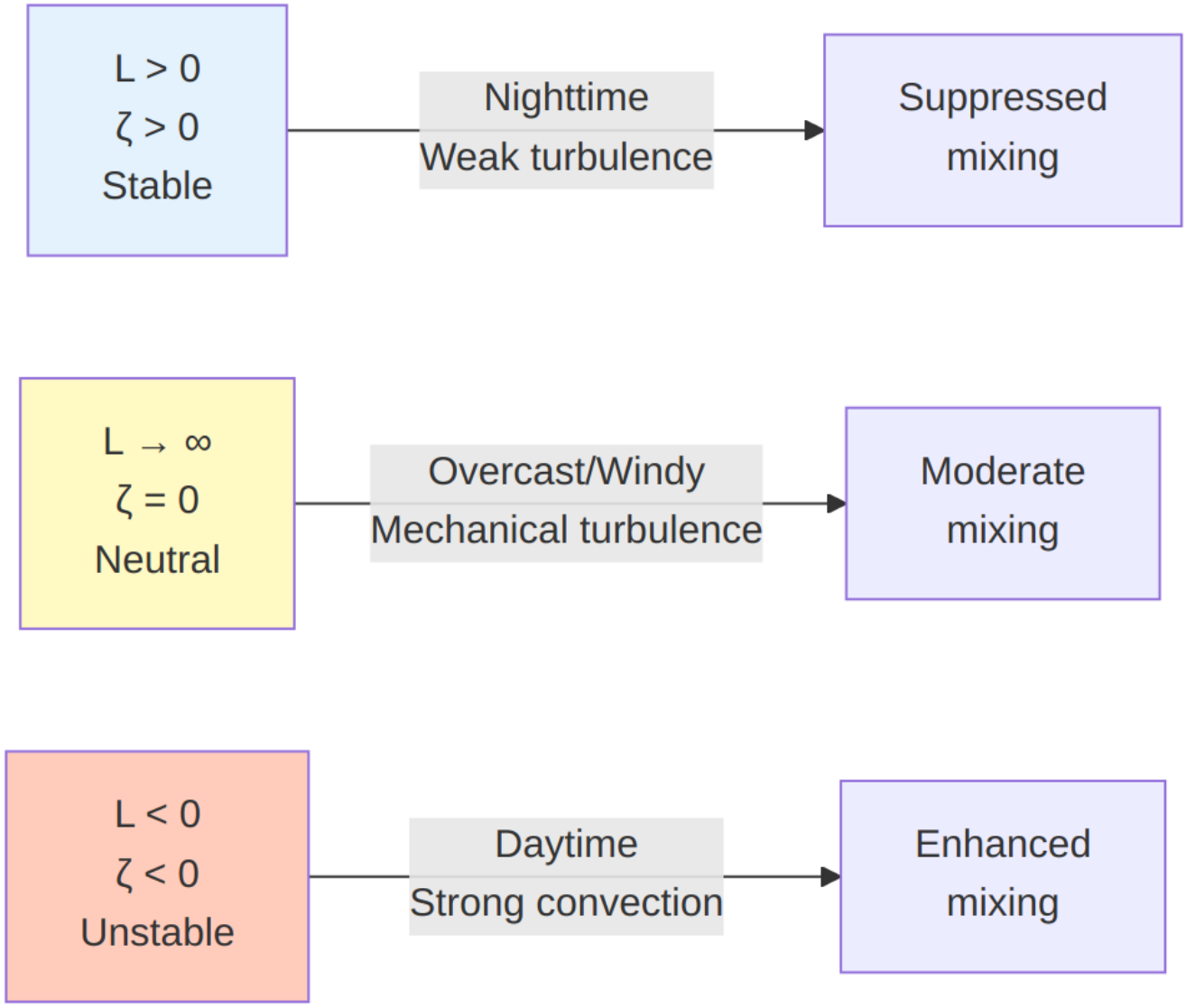
Atmospheric Stability Classes



Valid stability range used by the pipeline:

Condition	Threshold	Action
Very stable	$\zeta > 2$	K set to NaN — excluded from flux calculation
Very unstable	$\zeta < -5$	K set to NaN — excluded from flux calculation
Near-neutral	$\lvert\zeta\rvert \leq 0.01$	Neutral functions applied

5.1.5 Friction Velocity

The pipeline computes u_* from the 3-D sonic anemometer covariances:

$$u_* = (\overline{u'w'}^2 + \overline{v'w'}^2)^{1/4}$$

Double coordinate rotation is applied first in `db_process_hhour_sonic` to align the mean wind vector with the x-axis before computing turbulent covariances.

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Cup anemometer fallback

Code for a cup-anemometer-based u_* (iterative log-profile) exists in the MATLAB pipeline but is disabled. u_* is always derived from the sonic anemometer.

Gap-Filling Strategy for u*

When sonic-derived u_* is unavailable for a plot, the pipeline applies the following fallback hierarchy:

1. **Sonic u*** from the same plot (primary)
2. **Adjacent-plot u*** — used when the crop height difference satisfies $|h_{\text{diff}}|/h_{\text{mean}}<1$ (i.e., within 100 % of the mean crop height)
3. **No flux computed** — if no valid u_* source is available

5.1.6 Stability Functions (Businger-Dyer)

The pipeline uses the **Businger-Dyer formulation with coefficient 15**, implemented in both the flux function ϕ_h and its integrated form ψ_h.

Stability function φ_h

Condition	Formula
Stable (ζ > 0)	$\phi_h=1+4.7\zeta$
Neutral (|ζ| ≤ 0.01)	$\phi_h=1$
Unstable (ζ < 0)	$\phi_h=(1-15\zeta)^{-0.5}$

Integrated stability correction ψ_h

Used in the eddy diffusivity calculation:

Condition	Formula
Stable (ζ > 0)	$\psi_h=-4.7\zeta$
Neutral (|ζ| ≤ 0.01)	$\psi_h=0$
Unstable (ζ < 0)	$\psi_h=2\ln(\frac{1+x^2}{2})$, where $x=(1-15\zeta)^{0.25}$

5.1.7 Canopy Parameters

The MATLAB pipeline derives displacement height and roughness length from crop height h_c, using the following empirically calibrated factors for the Guelph agricultural sites:

Parameter	Symbol	Formula	Notes
Zero-plane displacement	d	$0.67\times h_c$	Standard agricultural value
Roughness length	z_0	$0.13\times h_c$	Site-calibrated; ~30 % larger than the classic 0.10 factor

Both parameters are defined per-site in the `*_init_all.m` initialization file and can be updated using dated epochs as crop height changes seasonally.

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5.1.8 Eddy Diffusivity

The integrated eddy diffusivity K is computed from the two sampling heights z_1 and z_2:

$$K=\frac{\kappa \cdot u_* \cdot \Delta z}{\ln(\frac{z_2-d}{z_1-d})-\psi_h(\frac{z_2-d}{L})+\psi_h(\frac{z_1-d}{L})}$$

where $\Delta z=z_2-z_1$.

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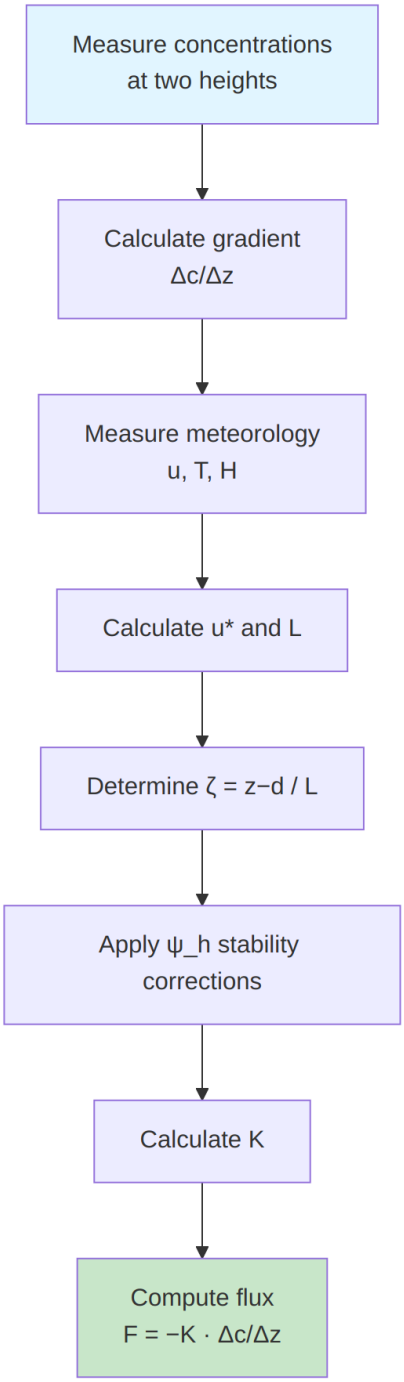
5.1.9 Flux Calculation

Step-by-Step Process



Practical Equation

For concentrations measured at heights z_1 and z_2:

$$F_c = -K \cdot \frac{c_2 - c_1}{z_2 - z_1}$$

In full integrated form:

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$$F_c = -\frac{\kappa \cdot u_* \cdot (c_2 - c_1)}{\ln(\frac{z_2-d}{z_1-d}) - \psi_h(\frac{z_2-d}{L}) + \psi_h(\frac{z_1-d}{L})}$$

Example Calculation

Simple Case: Neutral Conditions

Given:

- $c_1 = 410$ ppm at $z_1 = 0.5$ m
- $c_2 = 405$ ppm at $z_2 = 2.5$ m
- $u_* = 0.3$ m s⁻¹
- Neutral stability ($\psi_h = 0$, $\phi_h = 1$)
- $d = 0.2$ m (0.67 × 0.3 m crop)

$$K = \frac{0.40 \times 0.3 \times 2.0}{\ln(2.3/0.3)} = \frac{0.240}{2.037} = 0.118 \text{ m}^2\text{s}^{-1}$$

$$F_{CO_2} = -0.118 \times \frac{405 - 410}{2.0} = +0.295 \text{ ppm m s}^{-1}$$

5.1.10 Output Units

Gas	MATLAB Output Units
N_2O	ng N_2O-N m^{-2} s^{-1}
CO_2	µg CO_2 m^{-2} s^{-1}

Concentrations are stored internally in ppm (CO_2) or ppb/ppm (N_2O) and converted to flux units during the `db_calc_FG` stage.

5.1.11 Sensible Heat Flux Gap-Filling

When the sonic-derived sensible heat flux $\overline{w'T'}$ is unavailable, the pipeline falls back to:

1. **Sonic-derived H** from the same half-hour (primary)
2. **Neutral assumption** — sets $\psi_h = 0$ (equivalent to $L \to \infty$)

The neutral assumption introduces a small bias under stable/unstable conditions but is preferable to discarding the flux estimate entirely.

5.1.12 Quality Thresholds

The MATLAB pipeline applies a hierarchical QC scheme:

1. **Instrument diagnostics** — CSAT diagnostic flags, TGA detector stability
2. **Data completeness** — half-hours with > 50 % NaN values are rejected
3. **Stability range** — ζ > 2 or ζ < −5 → K = NaN
4. **Site-specific filters** — TGA pressure range, detector stability thresholds
5. **Meteorological conditions** — fetch, wind direction sectors (site-specific)
6. **Manual QC** — CSV-based filter files for known instrument problems

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Additional TGA-specific thresholds:

Parameter	Threshold	Action
Sample pressure	30–40 kPa (300–400 mb) range check	Flag / reject
ΔSample pressure between intakes	> 0.075 kPa	Flag for review
Outliers per half-hour	> 100 outlier samples	Reject half-hour
Zero or negative concentrations	Any	Convert to NaN

5.1.13 Advantages and Limitations

Advantages ✓

- **Compact footprint** — spatial averaging over shorter distances than EC
- **Multi-plot capability** — single TGA cycles through 4 plots simultaneously
- **Suitable for N_2O** — effective for low-flux gases where EC faces signal-to-noise challenges
- **Lower instrument requirements** — no ultra-fast response sensors needed

Limitations ✗

- **MOST assumptions required** — horizontal homogeneity, steady-state, surface layer
- **Stability dependent** — accuracy decreases for $\zeta > 2$ or $\zeta < -5$
- **Requires ancillary data** — u_*, L from sonic anemometer
- **Semi-empirical** — introduces additional uncertainty (~20–40 %)

Comparison with EC

Aspect	Flux-Gradient	Eddy Covariance
Measurement	Concentrations at 2+ heights	High-freq covariances
Minimum frequency	1–10 Hz	10–20 Hz
Theory base	MOST (semi-empirical)	Direct (fewer assumptions)
Footprint	Compact, height-dependent	Larger, stability-dependent
Stability range	Neutral–moderately unstable	All conditions
Multi-plot	Yes (TGA cycle)	No (single footprint)
Uncertainty	20–40 %	10–20 %

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5.1.14 Further Reading

Key References

1. **Businger et al. (1971)**. Flux-profile relationships in the atmospheric surface layer. *Journal of Atmospheric Sciences*, 28, 181–189. — *Stability functions used in the pipeline.*
2. **Brown, S.E., Wagner-Riddle, C., & Conrad, B. (2024)**. Low-power flux gradient measurements for quantifying the impact of agricultural management on nitrous oxide emissions. *Agricultural and Forest Meteorology*, 353, 110027. — *CWR Lab FG system design and validation.*
3. **Melman, E.A., et al. (2024)**. Increasing complexity in Aerodynamic Gradient flux calculations inside the roughness sublayer applied on a two-year dataset. *Agricultural and Forest Meteorology*, 355, 110107. — *FG method in the roughness sublayer.*
4. **Wagner-Riddle, C., et al. (2017)**. Globally important nitrous oxide emissions from croplands induced by freeze–thaw cycles. *Nature Geoscience*, 10(4), 279–283. — *N_2O application.*
5. **Foken, T. (2006)**. 50 years of the Monin-Obukhov similarity theory. *Boundary-Layer Meteorology*, 119, 431–447.

5.1.15 Next Steps

- TGA System — TGA100A operation and diagnostic variables
- Processing Pipeline — MATLAB script workflow
- Site Initialization — Configuring a new site

Quick Check

Why does the pipeline set K = NaN for ζ > 2 rather than using the stable formula?

Answer

Under very stable conditions (ζ > 2), MOST breaks down: turbulence becomes intermittent, the logarithmic profile assumption fails, and the Businger-Dyer stability functions are no longer reliable. Using the extrapolated formula would produce physically unrealistic diffusivities. Setting K = NaN explicitly excludes these periods rather than propagating a large systematic error into the flux record.

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5.2 TGA System Variables

This page documents the diagnostic variables and operational parameters of the **Campbell Scientific TGA100A Trace Gas Analyzer**, which forms the core of the flux-gradient measurement system at ON1.

> **Variable Context**
>
> All TGA variables are computed at **30-minute intervals** from high-frequency (10 Hz) measurements. These diagnostic parameters monitor system health and data quality.

5.2.1 Overview: TGA100A Operation

The TGA100A uses **tunable diode laser absorption spectroscopy** to measure N_2O and CO_2 concentrations in a temperature- and pressure-controlled sample cell.

Key Specifications

Parameter	Value
Target gases	N_2O and CO_2
Measurement frequency	10 Hz
Sample cell path length	153.08 cm
Operating pressure	50–80 mb
Sample flow range	500–2000 ml/min
Cooling system	Liquid nitrogen (LN_2)
Precision (N_2O)	< 0.3 ppb
Precision (CO_2)	< 0.1 ppm

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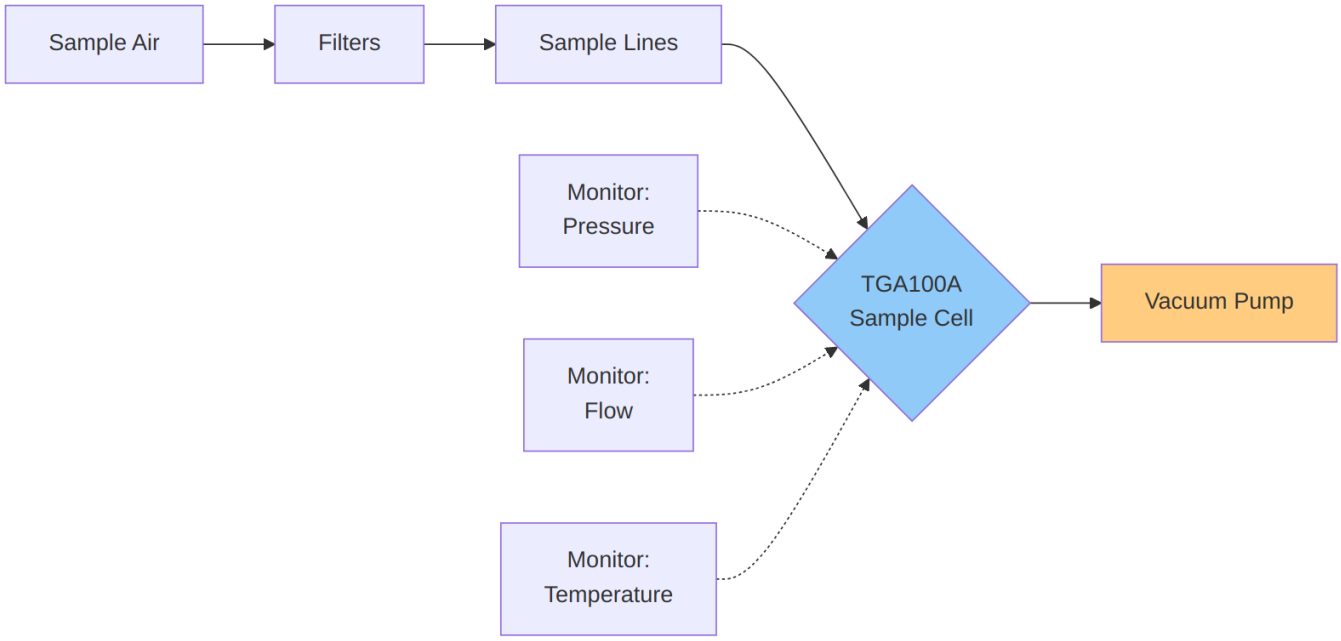

5.2.2 Measurement Cycle

Level Timing

The TGA cycles through intakes sequentially. Each level receives **60 seconds** of sampling, and an initial **omit period of 10–15 seconds** (configurable per plot in `*_init_all.m`) is discarded to allow the sample line to flush after a valve switch.

Stage	Duration	Notes
Reference gas (zero)	30 s	Baseline zero measurement
Per-level sampling	60 s	Valid window after omit period
Omit period (start of level)	10–15 s (configurable)	Flush time — discarded
Minimum valid level	≥ 12 s	Below this, half-hour is rejected

With a 4-plot system, **one complete cycle** (reference + 4 plots × 2 heights + reference) takes approximately 10 minutes, giving ~3 measurements per plot per 30-minute period.

Temporal coverage

Each plot receives roughly **1 flux estimate per 2 hours** in a 4-plot cycle (since each full cycle covers all plots, and fluxes are averaged over the 30-minute window).

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Tube Delay

The pipeline estimates tube delay using **sine-curve fitting to the CO_2 signal** during the nighttime window (20:30–07:30 local time). The estimated delay is stored in the init file and applied during structure processing.

Parameter	Value
Estimation method	Sine-curve fitting to CO_2 signal
Time window	Nighttime (20:30–07:30)
Typical delay (TGA100A)	2–6 seconds
Override mechanism	Override file for corrections

The `shiftDefault` parameter in `*_init_all.m` holds the current best estimate; it should be reviewed and updated when plumbing is modified or when new lag estimates become available.

Zero and Negative Concentrations

Any zero or negative concentration readings are **converted to NaN** before further processing. These represent non-physical values caused by electrical noise, ADC saturation, or sample cell contamination.

5.2.3 Pressure Variables

Sample Pressure

Units: mb (millibars)

The pressure inside the TGA sample cell directly controls measurement accuracy: it determines the molecular number density along the laser absorption path.

Operating range: 50–80 mb

Quality control requirements:

- Record daily and compare to 7-day moving average
- Flag if deviation > 10 % from baseline
- Investigate immediately if declining trend indicates filter clogging
- Check pump performance if sample pressure drops below 50 mb

> **Critical Parameter**
>
> Sample pressure directly affects measurement accuracy. Variations > 5 mb from typical operation should be investigated immediately.

QA/QC Thresholds:

Parameter	Min	Max	Flag Action
Sample Pressure	50 mb	80 mb	Check pump, filters, or plumbing
Acceptable operating range	300 mb	400 mb (30–40 kPa)	Flag if outside range

---

Delta Sample Pressure

Units: mb (millibars)

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Pressure difference between the upper and lower intakes in the multi-plot configuration.

Acceptable range: < 5 mb (< 0.075 kPa for each switching event)

Large ΔP between intakes indicates a flow imbalance that biases residence times and can affect gradient measurements.

Red Flag

ΔP > 5 mb between intakes indicates: - Blockage in one sampling line - Filter clogging at a specific intake - Bent or kinked tubing - Insect or debris in intake

Troubleshooting Workflow:

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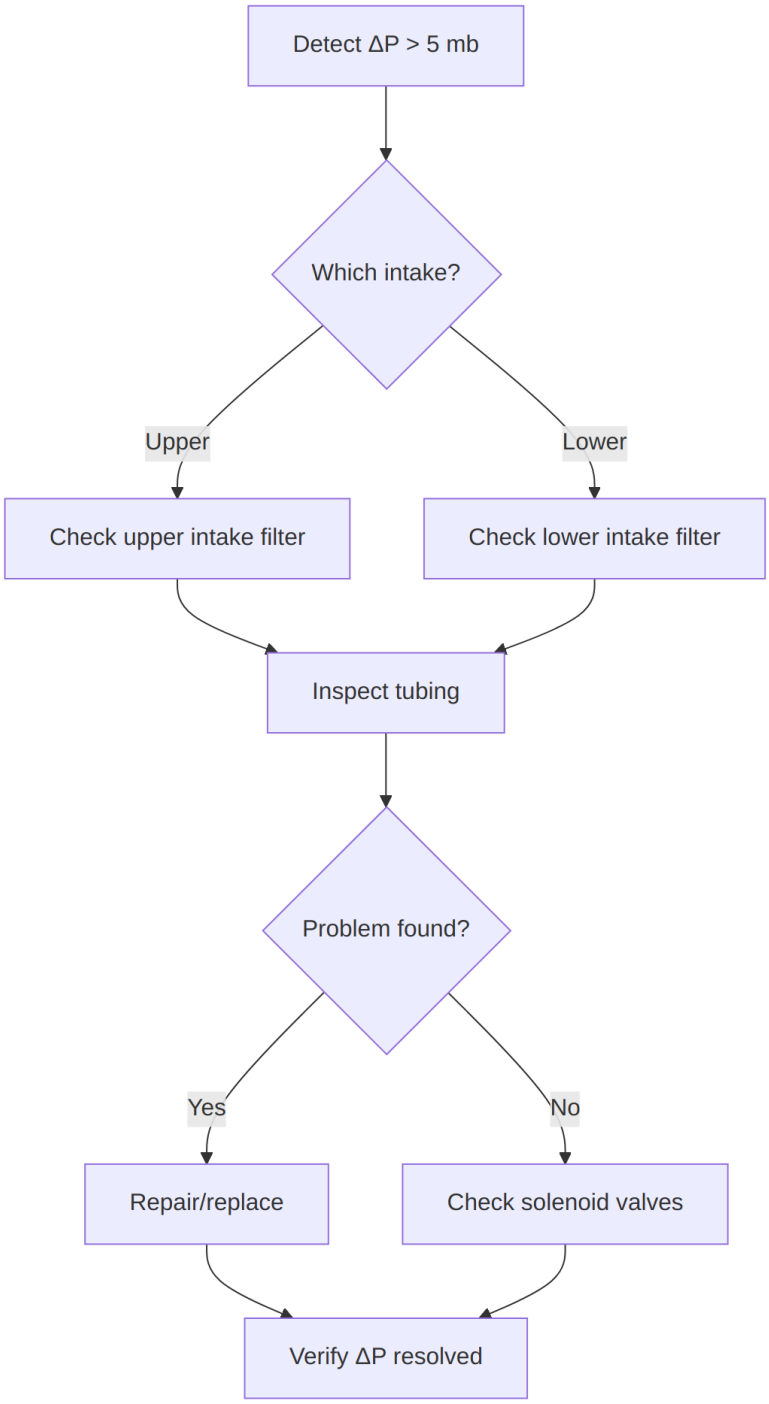

TGA Pressure

Units: mb (millibars)

Operating pressure inside the analyzer body (separate from sample cell pressure in some configurations). Must remain stable for consistent laser alignment and absorption.

Typical range: 50–80 mb

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Bypass Pressure

Units: mb (millibars)

Pressure in the bypass line of the multi-intake sampling system. The bypass maintains constant pressure while allowing sequential sampling from different intakes.

Sudden changes in bypass pressure indicate:

- Leak in sampling system
- Blocked intake line
- Pump malfunction
- Solenoid valve failure

5.2.4 Flow Variables

Sample Flow

Units: ml min^{-1}

Flow rate of air through the TGA sample cell.

Operating range: 500–2000 ml min^{-1} (site-specific optimum)

The flow rate is adjusted to achieve ~90 % laser transmittance — the balance between fresh sample replenishment and adequate absorption signal.

Quality control:

- Monitor for gradual decline (filter plugging — typical 10 % over 2–4 weeks)
- Investigate sudden drops immediately (leak or pump failure)
- Flag if flow changes > 5 % between consecutive 30-minute periods

---

Delta Sample Flow

Units: ml min^{-1}

Difference in flow rates between the upper and lower intakes.

Equal flow rates ensure comparable residence times, identical moisture conditioning, and unbiased concentration comparisons between intake heights. Persistent imbalance should be corrected by checking filters, verifying equal line lengths, or adding flow restrictors.

5.2.5 Quality Control: Outlier Threshold

The pipeline counts the number of 10 Hz samples flagged as outliers within each 30-minute period. If more than **100 outlier samples** are present in a half-hour, the entire half-hour is rejected.

5.2.6 Daily Monitoring Checklist

Morning System Check (5 minutes)

- [ ] Record sample pressure: _____ mb
- [ ] Record sample flow: _____ ml min^{-1}

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- [ ] Check ΔP between intakes: _____ mb
- [ ] Verify LN_2 level sufficient
- [ ] Review overnight data flags
- [ ] Check calibration gas pressure

End-of-Day Review (10 minutes)

- [ ] Calculate daily mean pressure: _____ mb
- [ ] Compare to 7-day baseline: _____ %
- [ ] Review flagged periods
- [ ] Update maintenance log
- [ ] Download data backup

5.2.7 Monitoring Dashboard

> 📹 *Video demonstration — see online documentation — available in the online documentation at the project site.*

Interactive dashboard: Real-time TGA system diagnostics

5.2.8 Reference Material

For complete TGA100A specifications and troubleshooting procedures, refer to:

- Campbell TGA100A Manual — Section 7.1.2 (Routine System Checks)
- Site-specific calibration protocols
- CWR Lab standard operating protocols and procedures

5.2.9 Next Steps

- FG Fundamentals — Theory and flux equations
- Processing Pipeline — MATLAB script workflow
- Troubleshooting — Common issues and solutions

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5.3 Flux-Gradient Processing Pipeline

This page documents the **Flux-Gradient (FG)** processing pipeline implemented in the CWR Lab MATLAB workflow. It focuses on **how the pipeline runs** and **what it produces**. Site-specific customisation is handled entirely through the site initialization file (`*_init_all.m`) and is documented separately in **Site Initialization**.

5.3.1 1. What FG Processing Does

The FG pipeline converts raw FG system data (TGA logger outputs and sonic/SmartFlux outputs) into:

- **Half-hour structured data** (MATLAB `.mat` structures)
- **Diagnostic vectors** (ASCII / MATLAB vector formats)
- **Flux-gradient calculations** (gradients, eddy diffusivity, fluxes)
- **Diagnostic plots** (TGA diagnostics, QC figures)
- **Archived outputs** (site-year products in standard output folders)

Supported gases: N_2O and CO_2

Output units:

Gas	Unit
N_2O flux	ng N_2O-N m^{-2} s^{-1}
CO_2 flux	µg CO_2 m^{-2} s^{-1}

5.3.2 2. Canonical Entry Points

The typical top-level runner is:

- `db_update_site(siteID, dateStr)` — run one site for a given day (or processing window)

A multi-site automation layer may call:

- `db_update_all(...)` — orchestration across sites and dates
- Windows Task Scheduler → BAT → Python → MATLAB wrappers (see the **Daily Automation** page)

The site adapter file is always loaded first via:

- `db_get_site_ini(siteID, dateStr)` → evaluates `dbIni = <SITEID>_init_all(dateStr)`

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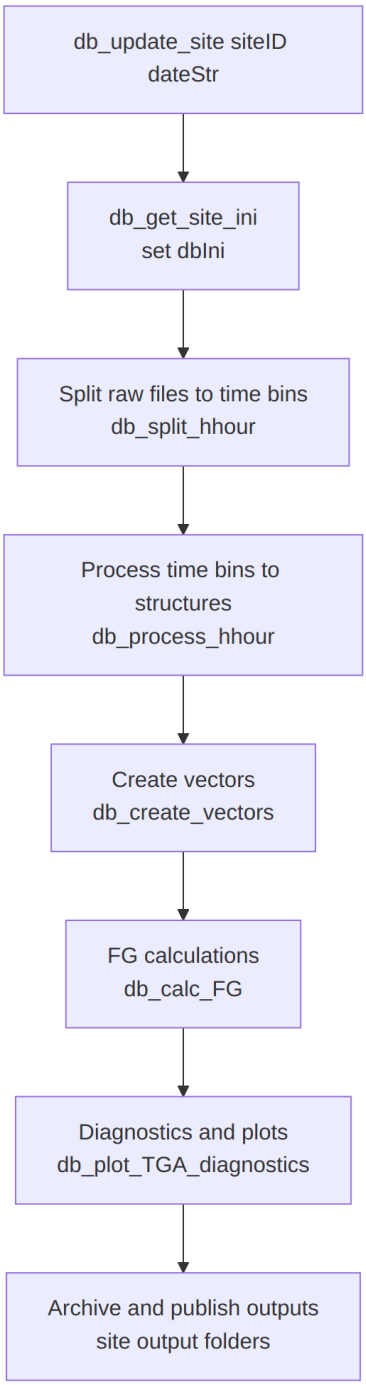
5.3.3 3. Pipeline Stages and Outputs



3.1 Split Raw Data → Half-Hour Files

Purpose: convert raw logger streams into consistent half-hour chunks aligned to the time step.

- Primary script: `db_split_hhour`
- Key `dbIni` fields: `hfPth`, `fileType`, `fileExt`, `headerLines`, `delimiter`, `outputDur`, `timeStep`

Output: half-hour chunk files (location set by site conventions)

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3.2 Process Half-Hours → MATLAB Structures

Purpose: parse half-hour chunks into standardised MATLAB structures.

- `db_process_hhour` — main entry point
- `db_process_hhour_sonic` — applies **double coordinate rotation** to align mean wind with the x-axis before computing turbulent covariances (u_*, $\overline{w'T'}$)
- `db_process_hhour_TGA` — processes TGA concentration data; applies tube delay correction using `shiftDefault`, discards omit period at level start (10–15 s configurable per plot)

Output: MATLAB `.mat` structure files under `structures/`

3.3 Structures → Diagnostic Vectors

Purpose: assemble continuous vector time series from per-half-hour structures.

- Primary script: `db_create_vectors`

Output: ASCII vectors and MATLAB vector files under `vectors/`

3.4 Flux-Gradient Calculations

Purpose: compute gradients, eddy diffusivity, and fluxes using vector products and site geometry.

- Wrapper: `db_calc_FG`
- Supporting scripts:
- `db_calc_grad` — concentration gradients
- `db_calc_rhoa` — air density (ρ_a)
- `db_calc_K` — eddy diffusivity using the integrated Businger-Dyer formulation
- `db_get_ENVCAN_clim` — fetches Environment Canada climate data for temperature and sensible heat flux gap-filling
- `db_make_heights` — resolves measurement heights from site geometry

Key constants applied:

Parameter	Value	Source
von Kármán constant κ	0.40	`Guelph_agromet_constants`
Gravity g	9.81 m s⁻²	Guelph latitude ~43.5 °N
Displacement factor	d = 0.67 × h_c	Default; overridable in `dbIni`
Roughness length	z_0 = 0.13 × h_c	Site-calibrated default
Stability valid range	−5 ≤ ζ ≤ 2	Outside → K = NaN
Stability functions	Businger-Dyer (coefficient 15)	See FG Fundamentals

Gap-filling hierarchy for missing inputs:

Input	Primary	Fallback 1	Fallback 2
u*	Sonic-derived	Adjacent plot (if $\lvert\Delta h\rvert/\bar{h} < 1$)	NaN (no flux)
Temperature	Sonic virtual temperature (T_v)	ENVCAN climate station	Default 288 K
Sensible heat flux H	Sonic-derived	Neutral assumption (ψ_h = 0)	—

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Output: calculation tables and products under `calculations/`

3.5 QC and Filtering

The pipeline applies a hierarchical quality control scheme:

1. Instrument diagnostics (CSAT flags, TGA detector stability)
2. Data completeness — reject if > 50 % NaN in the half-hour
3. Stability range — $\zeta > 2$ or $\zeta < -5 \rightarrow K = \text{NaN}$
4. TGA pressure check — operating range 50–80 mb; $\Delta P < 0.075$ kPa between intakes
5. Outlier count — > 100 outlier samples per half-hour → reject
6. Site-specific filters — wind direction, fetch sectors (defined in `*_init_all.m`)
7. Manual QC — CSV-based filter files for known instrument problems

---

3.6 Diagnostics and Plots

- Primary script: `db_plot_TGA_diagnostics(dbIni, fileNames, visibleOff)`
- Generates diagnostic images and figures under the site output tree

---

5.3.4 4. Testing a New Site Configuration

Before running a full year, validate in three steps:

1. **Split-only test** — can the pipeline find and split raw files for one day?
2. **Structures test** — can it produce valid half-hour MATLAB structures?
3. **Vectors + FG test** — can it generate vectors and produce at least one day of FG calculations?

If any step fails, review **Site Initialization** (instrument roster, parsing fields, `outputDur/timeStep`, plot mapping, and TGA manifold timing).

5.3.5 5. What Is Not Defined Here

Site-specific configuration is handled entirely in:

- **Site Initialization (`*_init_all.m`)** → see: *Flux-Gradient → Site Initialization*

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5.4 FG Site Initialization (`*_init_all.m`)

FG processing is designed so that **the MATLAB pipeline is generic** and **site-specific behavior lives entirely in** a single initialization function:

- `<SITE_ID>_init_all(dateStr)` → returns a `dbIni` structure

The processing pipeline loads this via `db_get_site_ini(siteID, dateStr)` and then uses `dbIni` everywhere downstream.

This page explains **how to adapt the system to a new site** by editing `*_init_all.m`.

5.4.1 1) Concept: the init file is a site adapter

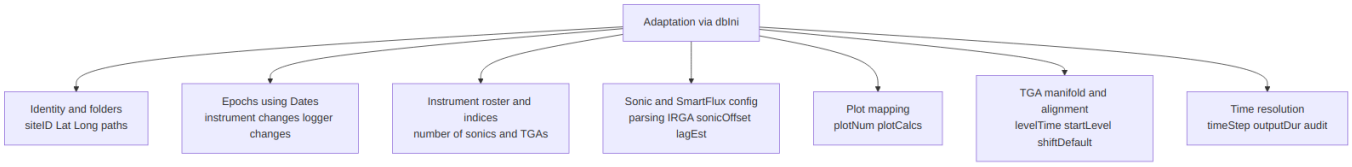
The init file is **not a static config**. It is a time-dependent function that:

- declares instruments, file formats, and paths
- maps instruments to plots and calculations
- encodes instrument changes over time using epochs
- controls parsing rules for TOA5 and SmartFlux inputs
- controls manifold timing and time alignment for TGA

Rule of thumb: if you are onboarding a new site, you should not modify the pipeline scripts first. Start by making the init file correct.

5.4.2 2) Adaptation levers

The init files across sites differ along a small number of consistent dimensions. These are the levers a user must understand.



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5.4.3 3) Required site-level fields (typical)

Most sites define the following top-level fields (names may vary slightly):

- **Identity**
- `dbIni.siteID`
- `dbIni.Lat`, `dbIni.Long`
- `dbIni.plotNum`
- **Paths**
- `dbIni.dbPth` — site root for raw and working products
- `dbIni.vecPth` — vectors output folder
- `dbIni.calcPth` — calculation products folder
- `dbIni.manPth` — manual measurements folder
- **Optional but common**
- `dbIni.emailList`
- profile cups and vanes metadata if used: `cupNum`, `cupPlotID`
- roughness and displacement defaults: `dHeightRatio`, `z0HeightRatio`, `d_est_bare`, `z0_est_bare`
- **Climate station linkage**
- `dbIni.ENVCAN_ID` or equivalent

5.4.4 4) Epochs: configuring changes over time

Most init files declare:

```
Dates(1) = datenum(YYYY,MM,DD,0,0,0);
```

Then instrument definitions are gated:

```
if datenum(dateStr,'yyyymmdd') > Dates(1)
    % define instruments for this epoch
end
```

Use epochs for: - instrument replacement - logger program changes - manifold or intake changes - major calibration changes requiring different defaults

Best practice: keep epochs small and explicit.

5.4.5 5) Instrument roster and indexing rules

Instruments are stored as:

- `dbIni.Instrument(i)`

Sites typically assign indices like:

- `nSonic_1 = 1`, `nSonic_2 = 2`
- `nTGA_1 = 3`

Hard rules: 1. Each index must be unique. 2. Keep a stable ordering: sonics first, TGAs after. 3. Do not re-use an index for a different instrument type in the same epoch.

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5.4.6 6) Sonic and SmartFlux configuration

Sonic-like instruments typically include:

- `Name`, `Type`, `Model`, `SerNum`
- `hfPth`, `strucPth`, `vecPth`, `dbPth`
- `Fs`

Parsing fields: - `dataOutputs` - `fileType` - `fileExt` - `delimiter` - `headerLines` - `tvCol` - `outputDur`

Processing fields: - `sonicOffset` - `IRGA` and `IRGAModel` - `lagEst` - `containsMet`

SmartFlux notes

SmartFlux exports often differ from TOA5 logger outputs:

- time vector may not exist as a tv column
- header structure differs
- output naming differs

Adaptation guidance - confirm header structure and delimiter - verify file cadence matches `outputDur` relative to `timeStep` - run a one day smoke test before scaling up

---

5.4.7 7) Plot mapping

Plot mapping controls which plots are associated with each sonic.

Key fields: - `plotNum` - `Instrument(i).plotCalcs`

Wrong mapping can produce physically wrong results without necessarily throwing an error.

5.4.8 8) TGA configuration

TGA blocks commonly include:

- parsing fields: `dataOutputs`, `fileType`, `delimiter`, `headerLines`, `tvCol`, `fileExt`, `outputDur`
- manifold and alignment fields:
- `levelTime`
- `startLevel`
- `omit`
- `lagEst`
- `shiftDefault`
- `corrIsotope`

Operational note: `shiftDefault` may change over time. Update it using epochs when needed.

5.4.9 9) Time resolution and outputDur

Many sites use half hourly timeStep 48.

If you switch to hourly timeStep 24, audit every instrument outputDur and re-run the smoke tests.

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5.4.10 10) New site adaptation checklist

Before editing the init file: - inventory instruments and file formats - map plot layout - document manifold timing and intake arrangement

Minimal run tests: 1. split-only test for one day 2. structures test for one day 3. vectors and FG calc test for one day

Full readiness: - climate station ID set - manual measurements available - QC diagnostics reviewed

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5.5 FG Troubleshooting (Quick Guide)

This page lists common failure modes when onboarding or modifying a site.

5.5.1 1) No files found or nothing splits

Likely causes: - wrong hfPth - wrong fileExt or fileType - headerLines mismatch causes the reader to reject data - date window does not satisfy the epoch condition using Dates

What to check: - confirm dbIni Instrument hfPth exists and contains expected raw files - confirm fileExt matches real files - confirm dateStr is after the epoch start

5.5.2 2) Split works but structures are missing or empty

Likely causes: - parsing mismatch delimiter header tv columns - dataOutputs names do not match actual output filenames - SmartFlux time handling not supported in your local processing scripts

What to check: - inspect one raw file and confirm delimiter and header assumptions - confirm output naming conventions

5.5.3 3) Vectors exist but FG calculations are wrong or missing

Likely causes: - incorrect plotCalcs mapping - missing manual measurements or heights - TGA alignment parameters wrong shiftDefault lagEst levelTime startLevel - climate station missing or wrong

What to check: - verify plot mapping against tower layout - verify manifold timing and start level - treat shiftDefault as tunable and update in epochs if it changed

5.5.4 4) Subtle bias or drift in gradients or fluxes

Likely causes: - sonicOffset incorrect - incorrect lag estimates - unmodeled change in tubing or manifold behavior over time

What to do: - confirm sonic orientation and apply correct offset - estimate lags using diagnostics and update lagEst and shiftDefault - introduce a new Dates epoch when configuration changes

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6. Complete Variable Glossary

This glossary provides comprehensive definitions for all diagnostic variables measured at the ON1 site. Variables are organized by measurement system.

Usage Notes

- All variables computed at 30-minute intervals
- Typical ranges are site-specific
- Establish baselines during commissioning
- Cross-reference with instrument manuals

6.1 Quick Navigation

- Flux-Gradient Variables
  - TGA Pressures
  - TGA Flows
  - TGA Temperatures
  - Concentrations
- Eddy Covariance Variables
  - Wind Components
  - Fluxes
  - Statistics
- Shared Variables

6.2 Flux-Gradient System Variables

6.2.1 TGA Pressures

Bypass Pressure

Symbol	P_bypass
Units	mb (millibars)
Definition	Pressure in the bypass line of the TGA sampling system
Purpose	Monitors pressure control in the multi-intake sampling system
Typical Range	Site-specific; depends on pump capacity and flow rates

Quality Control: - Should remain stable during operation - Changes indicate plumbing problems or leaks - Monitor for drift over time

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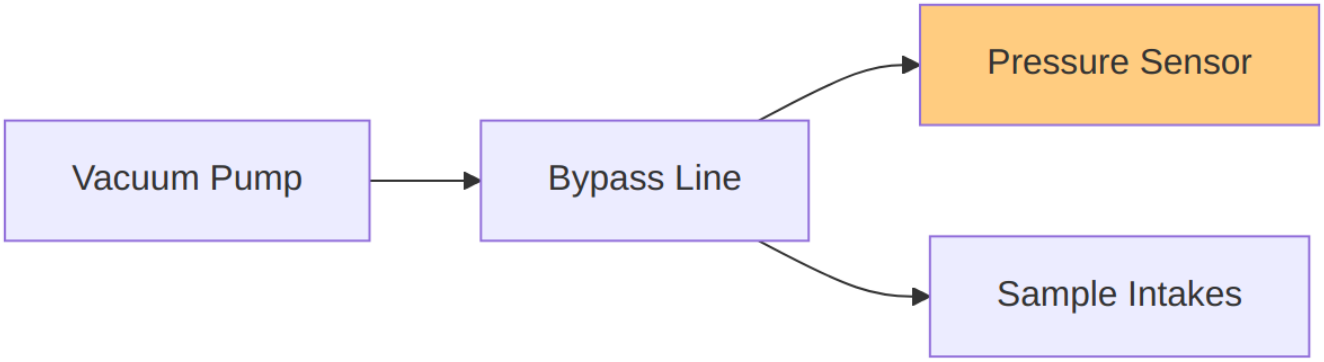

Sample Pressure

Symbol	P_sample
Units	mb (millibars)
Definition	Pressure in the TGA sample cell during measurement
Purpose	Critical for accurate concentration calculations; used in gas density conversions
Typical Range	40-100 mb (controlled by vacuum pump)

Critical Parameter

Sample pressure directly affects measurement accuracy. Changes > 5 mb from typical operation require immediate investigation.

Quality Control: - Record **daily** and compare to historical values - Changes indicate problems with flows/pressures in sampling system - Will decrease over time as filters become plugged

Daily Monitoring Protocol: 1. Record at start of day 2. Compare to 7-day moving average 3. Flag if deviation > 10% 4. Investigate declining trends

6.2.2 TGA Flows

Sample Flow

Symbol	Q_sample
Units	ml/min (milliliters per minute)
Definition	Flow rate of air through the TGA sample cell
Purpose	Determines residence time and affects measurement precision
Typical Range	500-2000 ml/min (site-specific)

Flow-Pressure Optimization:

The optimal flow rate balances: - Higher flow → Fresh sample, reduced memory effects - Lower flow → Better signal-to-noise, more absorption - Target: ~90% laser transmittance

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6.2.3 Gas Concentrations

N_2O Concentration

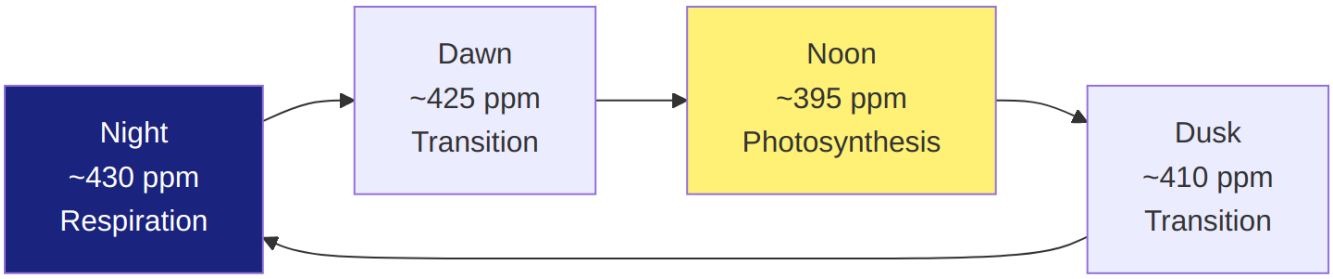
Symbol	[N_2O]
Units	ppb (parts per billion)
Definition	Nitrous oxide mole fraction in air sample
Purpose	Primary measurement for N_2O flux calculations
Typical Range	320-350 ppb background; higher over agricultural soils

Agricultural Context: - Background atmospheric: ~333 ppb - Enhanced by: Fertilizer application, soil disturbance - Temporal patterns: Episodic emissions, freeze-thaw events

CO_2 Concentration

Symbol	[CO_2]
Units	ppm (parts per million)
Definition	Carbon dioxide mole fraction in air sample
Purpose	For CO_2 flux calculations, comparison with EC system
Typical Range	400-450 ppm depending on time of day and season

Diurnal Pattern:



6.3 Eddy Covariance System Variables

6.3.1 Wind Components

u - Longitudinal Wind

Symbol	u, u'
Units	m/s
Definition	Horizontal wind component aligned with mean wind direction
Typical Range	-5 to +15 m/s (mean), ±5 m/s (fluctuations)

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v - Lateral Wind

Symbol	v, v'
Units	m/s
Definition	Horizontal wind component perpendicular to mean wind
Typical Range	Near 0 m/s (mean after rotation), ±3 m/s (fluctuations)

w - Vertical Wind

Symbol	w, w'
Units	m/s
Definition	Vertical wind component
Typical Range	~0 m/s (mean after rotation), ±2 m/s (fluctuations)

Coordinate Rotation

After proper coordinate rotation: - $\overline{v} \approx 0$ - $\overline{w} \approx 0$ - Only $\overline{u}$ should be non-zero

6.3.2 Turbulent Fluxes

CO_2 Flux

Symbol	F_CO_2 or F_c
Units	µmol/m²/s
Definition	Vertical turbulent flux of CO_2
Calculation	$F_{CO_2} = \overline{w'c'}$
Typical Range	-30 to +15 µmol/m²/s

Sign Convention: - **Negative**: Downward flux (uptake by vegetation) - **Positive**: Upward flux (respiration)

Typical Daily Pattern:

Time	Typical Flux	Process
06:00	+5 µmol/m²/s	Early respiration
12:00	-20 µmol/m²/s	Peak photosynthesis
18:00	-5 µmol/m²/s	Declining uptake
24:00	+10 µmol/m²/s	Nighttime respiration

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Sensible Heat Flux

Symbol	H
Units	W/m²
Definition	Vertical turbulent flux of sensible heat
Calculation	$H = \rho c_p \overline{w'T'}$
Typical Range	-50 to +400 W/m²

Where: - ρ = air density (kg/m³) - c_p = specific heat of air (1005 J/kg/K) - T = air temperature (K)

Latent Heat Flux

Symbol	LE or λE
Units	W/m²
Definition	Vertical turbulent flux of latent heat (evapotranspiration)
Calculation	$LE = \lambda \overline{w'\rho_v'}$
Typical Range	0 to +500 W/m²

Where: - λ = latent heat of vaporization (2.45 MJ/kg) - ρ_v = water vapor density (g/m³)

6.3.3 Turbulent Statistics

Friction Velocity

Symbol	u*
Units	m/s
Definition	Measure of turbulent intensity in surface layer
Calculation	$u_* = [(\overline{u'w'})^2 + (\overline{v'w'})^2]^{1/4}$
Typical Range	0.1 to 1.0 m/s

Importance: - Characterizes turbulent mixing - Critical for flux-gradient calculations - Quality control parameter (u* threshold) - Footprint calculations

Quality Threshold: - Nighttime fluxes typically filtered when $u_* < 0.1$ m/s (site-specific)

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Obukhov Length

Symbol	L
Units	m
Definition	Characteristic length scale of atmospheric stability
Calculation	$L=-\frac{u_*^3\overline{T}}{\kappa g\overline{w'T'}}$
Typical Range	-∞ to +∞

Interpretation:

L Value	Stability	Typical Time	Mixing
L < 0	Unstable	Daytime	Enhanced
L → ±∞	Neutral	Cloudy/Windy	Moderate
L > 0	Stable	Nighttime	Suppressed

6.4 Shared Meteorological Variables

6.4.1 Temperature

Air Temperature

Symbol	T_air
Units	°C or K
Definition	Ambient air temperature at measurement height
Measurement	Thermistor or thermocouple in aspirated shield

Sonic Temperature

Symbol	T_sonic or T_s
Units	°C or K
Definition	Temperature derived from sonic anemometer (speed of sound)
Relationship	$T_s=T(1+0.51q)$ where q is specific humidity

Virtual Temperature

Sonic temperature includes moisture effects and must be corrected to obtain actual air temperature.

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6.4.2 Humidity

Relative Humidity

Symbol	RH
Units	%
Definition	Ratio of actual to saturation vapor pressure
Typical Range	20-100%

Specific Humidity

Symbol	q
Units	g/kg
Definition	Mass of water vapor per mass of moist air
Calculation	$q = 0.622 \frac{e}{P - 0.378e}$

Where: - e = vapor pressure (Pa) - P = total pressure (Pa)

6.4.3 Radiation

Net Radiation

Symbol	R_n
Units	W/m²
Definition	Net radiative flux at surface
Calculation	$R_n = SW_{in} - SW_{out} + LW_{in} - LW_{out}$

Components: - SW_in: Incoming shortwave - SW_out: Reflected shortwave - LW_in: Incoming longwave - LW_out: Outgoing longwave

6.5 Using This Glossary

6.5.1 For New Users

1. **Start with basics**: Understand main flux variables first
2. **Learn QA/QC**: Focus on quality control parameters
3. **Cross-reference**: Use with method chapters for context

6.5.2 For Data Analysis

1. **Variable selection**: Choose appropriate variables for analysis
2. **Units**: Verify unit consistency
3. **Quality flags**: Apply appropriate filtering
4. **Typical ranges**: Use to identify outliers

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6.5.3 For Troubleshooting

1. **Check diagnostics**: Start with pressure/flow variables
2. **Compare typical ranges**: Identify deviations
3. **Time series analysis**: Look for trends
4. **Cross-validation**: Compare FG and EC when available

6.6 Measurement Schedule

Variable Type	Measurement Frequency	Output Interval
TGA diagnostics	10 Hz	30 minutes
EC raw data	10 Hz	Stored as 10 Hz
EC fluxes	Computed from 10 Hz	30 minutes
Meteorology	1 Hz or slower	30 minutes

6.7 References

For complete details on variables and measurement principles:

- **Introduction**: Site Overview
- **EC Variables**: EC Variables Guide
- **FG Variables**: FG Variables Guide
- **TGA System**: TGA System Details
- **References**: Scientific Literature

This glossary is based on the ON1 Variables Glossary Documentation (Version 1.0, December 2025)

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7. References and Further Reading

7.1 Instrument Manuals

All instrument manuals are available in the project files:

7.1.1 Campbell Scientific TGA100A

Campbell TGA Manual

Key Sections:

- Section 5.1: Measurement Specifications
- Section 7.1.2: Routine System Checks
- Appendix A: Gas Configuration
- Appendix E: Calibration Procedures

7.1.2 Campbell Scientific CSAT-3

Campbell CSAT-3 Manual

Key Sections:

- Section 2: Specifications
- Section 11.2.2: Wind Offset Testing
- Appendix B: Serial Communications

7.1.3 LI-COR Li-7500

LI-COR Li-7500 Manual

Key Sections:

- Section 3: Setup and Operation
- Section 4: Calibration
- Section 5: Maintenance
- Appendix A: Specifications

7.1.4 Campbell Scientific IRGASON

Product webpage: https://www.campbellsci.ca/irgason

7.2 Scientific Literature

7.2.1 Flux-Gradient Method

Baldocchi, D.D., Hicks, B.B., & Meyers, T.P. (1988). Measuring biosphere-atmosphere exchanges of biologically related gases with micrometeorological methods. *Ecology*, 69(5), 1331-1340.

Meyers, T.P., et al. (1996). A comparison of summertime water and CO_2 fluxes over rangeland for well watered and drought conditions. *Agricultural and Forest Meteorology*, 89, 87-105.

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Wagner-Riddle, C., et al. (2007). Flux of nitrous oxide from agricultural fields: Measurement and flux-gradient approach. *Journal of Geophysical Research*, 112, D24112.

Zhao, J., et al. (2019). An evaluation of the flux-gradient and the eddy covariance method to measure CH_4, CO_2, and H_2O fluxes from small ponds. *Agricultural and Forest Meteorology*, 275, 255-264.

7.2.2 Monin-Obukhov Similarity Theory

Monin, A.S., & Obukhov, A.M. (1954). Basic laws of turbulent mixing in the surface layer of the atmosphere. *Tr. Akad. Nauk SSSR Geophiz. Inst*, 24, 163-187.

Businger, J.A., et al. (1971). Flux-profile relationships in the atmospheric surface layer. *Journal of Atmospheric Sciences*, 28, 181-189.

Högström, U. (1988). Non-dimensional wind and temperature profiles in the atmospheric surface layer: A re-evaluation. *Boundary-Layer Meteorology*, 42, 55-78.

Foken, T. (2006). 50 years of the Monin-Obukhov similarity theory. *Boundary-Layer Meteorology*, 119, 431-447.

Paulson, C.A. (1970). The mathematical representation of wind speed and temperature profiles in the unstable atmospheric surface layer. *Journal of Applied Meteorology*, 9, 857-861.

7.2.3 Eddy Covariance Method

Baldocchi, D.D. (2014). Measuring fluxes of trace gases and energy between ecosystems and the atmosphere - the state and future of the eddy covariance method. *Global Change Biology*, 20, 3600-3609.

Webb, E.K., Pearman, G.I., & Leuning, R. (1980). Correction of flux measurements for density effects due to heat and water vapour transfer. *Quarterly Journal of the Royal Meteorological Society*, 106, 85-100.

Moncrieff, J.B., et al. (1997). A system to measure surface fluxes of momentum, sensible heat, water vapour and carbon dioxide. *Journal of Hydrology*, 188-189, 589-611.

7.2.4 Data Processing and Quality Control

Mahrt, L. (1998). Flux sampling errors for aircraft and towers. *Journal of Atmospheric and Oceanic Technology*, 15, 416-429.

Griffis, T.J., et al. (2003). Ecophysiological controls on the carbon balances of three southern boreal forests. *Agricultural and Forest Meteorology*, 117, 53-71.

Foken, T., & Wichura, B. (1996). Tools for quality assessment of surface-based flux measurements. *Agricultural and Forest Meteorology*, 78, 83-105.

7.2.5 Agricultural Applications

Amiro, B.D., et al. (2006). Carbon, energy and water fluxes at mature and disturbed forest sites, Saskatchewan, Canada. *Agricultural and Forest Meteorology*, 136, 237-251.

Phillips, R.L., et al. (2007). Soil-atmospheric exchange of nitrous oxide in Canadian agricultural soils. *Journal of Geophysical Research*, 112, D03304.

7.2.6 Turbulence and Boundary Layer

Kaimal, J.C., & Finnigan, J.J. (1994). *Atmospheric Boundary Layer Flows: Their Structure and Measurement*. Oxford University Press.

Stull, R.B. (1988). *An Introduction to Boundary Layer Meteorology*. Kluwer Academic Publishers.

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7.3 Theses and Dissertations

7.3.1 Site-Specific Research

Pow, P. (2024). Carbon dioxide, water vapour, and nitrous oxide fluxes over an intensively managed forage crop in the Lower Fraser Valley in British Columbia, Canada. PhD Thesis, University of British Columbia.

View Thesis

Relevant Chapters:

- Chapter 2: Climate and flux measurements (methods, pp. 19-35)
- Chapter 5: FG method evaluation (pp. 117-147)

7.4 Online Resources

7.4.1 Educational Materials

- **Micrometeorology Course** - University of British Columbia
  Advanced micrometeorology and flux measurements
- **FluxNet Community**
  https://fluxnet.org/
  Global network of flux measurement sites
- **Eddy Pro Software Documentation**
  https://www.licor.com/eddypro
  Free EC data processing software

7.4.2 Data Processing Tools

- **EddyPro** (LI-COR)
  Free, open-source EC processing software
- **TK3** (University of Bayreuth)
  Turbulence and flux analysis package
- **PyFluxPro** (Python)
  Python-based flux data processing

7.5 Standards and Best Practices

Aubinet, M., et al. (2012). *Eddy Covariance: A Practical Guide to Measurement and Data Analysis*. Springer.

Burba, G. (2013). *Eddy Covariance Method for Scientific, Industrial, Agricultural and Regulatory Applications*. LI-COR Biosciences.

7.6 Contact Information

7.6.1 Technical Support

Campbell Scientific Technical Support
Email: support@campbellsci.com
Phone: 1-435-227-9120

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LI-COR Technical Support
Email: envsales@licor.com
Phone: 1-800-447-3576

7.6.2 Academic Contacts

Site Manager: [To be added]
Lab Supervisor: [To be added]
Data Manager: [To be added]

7.7 Revision History

Version	Date	Changes
1.0	Dec 2025	Initial compilation from instrument manuals
		Incorporated methods from Pow (2024) thesis
		Based on published literature

Next Updates:

- Site-specific calibration values
- Local measurement heights
- Footprint analysis results
- Site photographs and diagrams

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